Carrying Capacity

Island Resources Officer · Vellan Island

15 DAYS · 49 STOPS · GRADE 12 · CARRYING

THE OPENING CARD

There are 11 months of fresh water left under this island. In a fortnight the council votes on a second ferry run each day, which brings more people and no more rain. You are the island's resources officer, and you sign the water licence, the fish quota and the tip. In 15 days you write the Vellan conditions the vote is taken on. 91 people live here, and the school shuts if the 19 on its roll drop to 12.

Ninety-one people and one of everything

BEFORE THE DAY — TRIAL

Walk the harbour end before you sign anything

The office, the waterworks and the common all stand inside four hundred metres of the berth, so walk that end once and see which is which.

PLAN CARD 159W

Tuesday. The boat that brought you takes the outgoing officer off at four. Ines Calloway, the harbourmaster, has the landings book open on the desk. The page shows 168 tonnes this year against 214 five years ago. She wants that figure in the council pack before the ferry paper is drafted. Tomas Berhane, the reef ecologist, says the same page has been quoted three times, with a different period attached each time. The Vellan conditions are four rates of that kind: water pumped, fish landed, waste buried, boats run. Each one has to carry the period it was written from. Today you settle what each of the island's four books measures, and which of them can go in front of a vote. Torr Vey grazes four shares on the common and has already asked which book his limit comes out of. A figure with no year on it comes back in three years as a fact nobody can trace.

BRIEFING 17W

The handover is four books and a licence, and no two of them cover the same year.

WHAT YOU DECIDE 13W

Establish what the island's budgets actually are before anybody argues about the ferry.

1.1 Down from what, over how long

HARBOUR OFFICE · A ROOM · BALLPARK

SCENE 36W

Calloway has the landings book open at two pages. The west ground landed 214 tonnes in the season five years ago and 168 tonnes last season. The ferry paper quotes a fall without saying from when.

GUIDE 62W

A change is only a claim once it says what it is a change from. Take the difference between the two seasons and divide it by the earlier one. Dividing by the later one answers a different question, and gives a different number. Then check what period the ferry paper attaches to it, and whether that is the period you just used.

ASK 10W

Estimate the percentage fall in landings across the five seasons.

BEHIND THE BUTTON 175W

Why the denominator is the earlier value. A percentage change measures how much of the original has gone, so the original is what it is a fraction of. Dividing by the new value gives a real number about something else. It gives how much the smaller figure would have to grow to get back. The two are never the same.

What a percentage hides. Twenty-one per cent of 214 tonnes is 46 tonnes, and 46 tonnes is either a crisis or a rounding error depending on the stock it came out of. A percentage travels well between contexts and loses the size of the thing it is about, which is why a council paper carries both.

Why the period matters more than the arithmetic. The same two numbers describe three different things, depending on what is put beside them. A five-year fall. An annual rate of a little over four per cent. Or a single bad season. Only one of those is what the book records, and the other two are how it gets misquoted.

TAKES AS READ

the landings book records whole seasons rather than calendar years

VERDICT 101W

The difference is 46 tonnes and the comparison is against the 214 the fall started from, which gives 21.5 per cent. Dividing by last season's 168 gives 27 per cent instead. That number answers how much the catch would have to rise to recover. It does not answer how much of it has gone. Both are arithmetic; only one is the claim the paper is making. The period is the part that travels badly. Twenty-one per cent over five seasons and twenty-one per cent in a year are the same figure describing two different fisheries, and the book records the first.

TAKEAWAY 16W

A change is a fraction of where it started, and it means nothing without its period.

1.2 How slow the store is

WATERWORKS · A PERSON · BALLPARK

SCENE 27W

The aquifer holds an estimated 2.6 million cubic metres. Recharge is the 294,000 cubic metres a year you measured, and the borehole is licensed to take 340,000.

GUIDE 58W

Divide what is held by what moves through each year. That is the residence time — how long the store takes to replace itself. It is the number that says how fast the aquifer responds to anything, in both directions. How long an over-draw takes to show up. And how long a corrected licence takes to fix it.

ASK 8W

Estimate the residence time of the island's aquifer.

BEHIND THE BUTTON 175W

Why a slow store is dangerous rather than safe. A big buffer absorbs a mistake without showing it, so the evidence of over-abstraction arrives years after the licence that caused it. By the time the chloride is unambiguous, the decision being questioned is a decade old and the people who made it have left.

Why the recovery takes the same time. The clock is set by the ratio of the store to the flow, and the flow does not change when a cap is imposed. Capping abstraction tomorrow does not restore the balance tomorrow. It restores it over the same nine-year turnover. That is what a council expecting a quick result has to be told.

What the number rests on. The stock estimate is the weakest term. It comes from a 1998 geophysical survey with a range of about thirty per cent. The recharge, by contrast, is now measured. A residence time quoted to the year from a stock known to a third is a false precision, and the honest form is "roughly nine years".

TAKES AS READ

the aquifer stock estimate is of the same order as the 1998 survey · an aquifer, and what abstracting faster than recharge does to it — taken as read

VERDICT 114W

Two point six million cubic metres over 294,000 a year is a little under nine years. That single number explains the shape of the whole argument. A licence written from a mainland table in 1998 has been over-drawing by about fifteen per cent. A store that replaces itself once a decade absorbed it quietly. It went on absorbing it until the July chloride became undeniable, which is now. It also disposes of the hope in the room: capping abstraction does not restore the balance this season, it restores it across another nine years. The stock is the weak term, known to about a third, so the honest answer is "roughly nine" rather than 8.8.

TAKEAWAY 17W

Residence time is stock over flow, and it sets how slowly a store both fails and recovers.

1.3 Nine inches over rock

COMMON OFFICE · A ROOM · PROTOCOL

SCENE 51W

Dan Pike, the common grazing officer, has a pit dug at the top of the common. The pit shows nine inches of dark topsoil, two inches of paler material under it, and rock. Ewan Hollis, who runs the market garden, wants another acre in vegetables if the ferry brings a market.

GUIDE 63W

A soil profile is read as layers, and the top one is the one that grows things. Look at how deep it is, what is under it, and how fast water leaves it. Then ask what a crop takes out each year. Ask whether nine inches of soil over rock can replace it. On an island the replacement has to be made here.

ASK 11W

Match each layer to what it tells you about growing here.

BEHIND THE BUTTON 167W

What the horizons are. A dark surface layer rich in organic matter, a paler layer below it where minerals have washed down, and then parent rock. Deep soils have more between the two; this one has almost nothing, which is normal for a grazed maritime island and limits what can be asked of it.

Why texture decides as much as depth. Sand drains fast and holds little; clay holds water and air badly; loam is the mixture that does both. Vellan's is a sandy loam over rock, which drains quickly. That is good in a wet spring. It is also the reason a dry August is felt in the polytunnels within a fortnight.

What the common's history has already done. Two hundred years of grazing has kept organic matter cycling in place through dung and has stopped anything woody establishing. Hollis's two acres are the only deep soil on the island, and every inch of it was built out of seaweed and what the tip would release.

TAKES AS READ

the pit at the top of the common is representative of the grazed ground

VERDICT 109W

A profile is read as a set of limits rather than as a description. Nine inches of topsoil is what carries the organic matter, the nutrients and nearly all the roots, so it is the whole of the productive soil here. The pale layer beneath is leached mineral material and adds little. The rock at eleven inches means there is no deep store to fall back on. The twenty-minute drainage says the texture is sandy. So a dry month is felt quickly rather than buffered. Together those four say the common can carry grazing and cannot carry another acre of vegetables without soil being built first, which takes years.

TAKEAWAY 20W

A thin soil over rock limits what can be grown, because the replacement has to be made on the island.

END OF THE DAY 17W

Every environmental number is a stock or a flow, and it is worth nothing without its period.

DAY 2 OF 15

One deposit, four standing orders

PLAN CARD 118W

Wednesday. The abstraction licence is the island's permit to pump water. The renewal has to be lodged before the ferry paper is drafted. The ferry will need water too, and the paper is quiet about that. Mairead Sorley, the water engineer, has 14 years of her own rain readings. She also has a meter on the borehole. Nobody has ever asked her to put the two together. Calloway wants a figure she can defend at the council. Berhane wants the licence capped at the recharge. Today you measure the deposit, count the standing orders, and say how long the store would last if the rain stopped. Get the deposit wrong and the cap means nothing for ten years.

BRIEFING 17W

The abstraction licence is up for renewal and nobody has measured the recharge since it was written.

WHAT YOU DECIDE 15W

Measure what the island's aquifer is actually given each year, and what is taken out.

2.1 What the rain leaves behind

WATERWORKS · A ROOM · BALLPARK

SCENE 37W

Sorley has fourteen years of rain readings. They average 1.05 metres a year. The catchment above the borehole is 1.4 square kilometres, and the original licence says a fifth of the rain gets down to the aquifer.

GUIDE 79W

Rain that lands is not water you can have. Some runs off the rock to the sea. Most of the rest goes back up through the turf. What is left soaks down to the store the borehole draws from. Multiply the depth of rain by the area it falls on to get a volume, then take the fraction that actually gets down there. Watch the units: a depth in metres times an area in square metres is cubic metres.

ASK 7W

Estimate the annual recharge to the aquifer.

BEHIND THE BUTTON 170W

Why the coefficient is so small. On Vellan the rock is close to the surface, the turf is short and the wind is constant. So evaporation and transpiration take the largest share, and runoff takes most of the rest. A fifth reaching the aquifer is normal for a thin maritime soil and would be wrong for a deep sand.

What the number is and is not. It is an annual average over a catchment, which means it is right about a decade and silent about August. The store buffers the difference between a wet winter and a dry summer. The licence has to be written against the average. The tanks are sized against the worst month.

Why nobody has measured it. The licence was written in 1998 from a mainland table, and the figure has been repeated ever since. Sorley's fourteen years of readings are the first local evidence. They are about eight per cent lower than the table. That matters, because the licensed abstraction was set from the table.

TAKES AS READ

the fourteen-year mean rainfall is representative of the next decade · an ecosystem as a budget: what comes in, what leaves, what stays — taken as read

VERDICT 103W

A depth of rain over an area is a volume. Only the fifth that gets underground is available. 1.05 metres over 1.4 square kilometres is 1.47 million cubic metres of rain. Of that, 294,000 reaches the store. The licence allows 340,000. That gap is the whole of the day, and it is not a modelling subtlety. The licence was written from a mainland recharge table. Sorley's readings are the first local measurement. Using the 0.8 instead gives 1.18 million, which is the water that never arrives. It is the largest number on the board, and it belongs to the sea and the sky.

TAKEAWAY 17W

Recharge is rain times area times the fraction that gets underground, and it is an annual average.

2.2 One more animal each

COMMON OFFICE · A PERSON · CHOICE

SCENE 28W

Pike's count puts 512 sheep on a common assessed at 430. Nine graziers hold forty-one shares between them, and not one of them is over their own share.

GUIDE 58W

Work out where the extra animals came from before deciding what rule answers it. Look at who gains from one more animal and who pays for it, and whether any individual grazier has broken anything. Then pick the rule that changes the arithmetic each grazier faces, rather than the one that describes what you wish they would do.

ASK 8W

Which rule actually answers what the count shows?

BEHIND THE BUTTON 170W

Why nobody has to cheat for this to happen. One more animal is worth having to the grazier who owns it, and the cost — slightly worse grazing — is spread across all nine. Each individual decision is rational and the sum of them exceeds the common's capacity. That is the structure, and moralising about it changes nothing.

Why shares alone did not hold it. The shares were set as fractions of a total that was never inspected. So nine graziers each within their share can still sum past the limit as the total drifts. A rule expressed as a proportion needs the total to be counted; a rule expressed in animals does not.

What the three real options do. A hard stocking limit with inspection changes what each grazier may do. A charge per animal above a threshold changes what each animal costs them. Dividing the common into private parcels internalises the cost entirely and ends the common — which is a real answer and has its own price.

TAKES AS READ

Pike's count is of animals on the common on one day · carrying capacity, and what happens on the way past it — taken as read

VERDICT 102W

Nobody here has broken their share. So the failure is in the rule rather than in the graziers. Shares are proportions of an uninspected total, and nine people inside their proportion can still sum past 430. A limit expressed in animals and actually counted fixes precisely that, and publishing the count is what makes it enforceable between neighbours who have to live together. A reminder changes no arithmetic. Cutting every share proportionally keeps the same uncounted structure and simply moves the number. Banning new graziers protects the nine who are already over and does nothing about the 512 animals on the ground.

TAKEAWAY 20W

The commons fails because the gain is private and the cost is shared, and only the arithmetic can be changed.

2.3 Ninety-one, and how many in July

HARBOUR OFFICE · A ROOM · CHOICE

SCENE 28W

Calloway has the resident register at ninety-one and the ferry manifests for last July. The manifests show 340 day visitors and 61 people staying overnight across the month.

GUIDE 60W

A population is not the same as a load. Ask what each kind of person actually draws from the island's budgets — water, waste, power — and for how long. A day visitor and a resident are not the same number of anything. Adding them as though they were is how the ferry paper reaches a figure that looks harmless.

ASK 11W

Which comparison tells the council what July actually costs the island?

BEHIND THE BUTTON 154W

What a footprint measures. The land and water needed to supply what somebody consumes and absorb what they discard. On an island the interesting part is not the total. It is the share of that total which has to be met locally. Water and waste cannot be imported or exported cheaply. Power partly can.

Why day visitors and overnight visitors are different quantities. A day visitor uses a toilet, buys lunch and leaves; an overnight visitor showers, washes and produces a day's waste. The water difference between the two is roughly tenfold, and the ferry paper's headline counts them together.

Why peak matters more than average here. Three hundred and forty visitors spread over a year is invisible in every budget. The same number in July arrives when the borehole is at its lowest. The tanks are already being managed by then. July is the one month the island's spare capacity is not spare.

TAKES AS READ

the register and the manifests both cover the same July · carrying capacity, and what happens on the way past it — taken as read

VERDICT 102W

The budgets feel person-days, not arrivals, and they feel them in the month they happen. Sixty-one overnight visitors across July is a few hundred visitor-nights. Each of those nights draws something like a resident's daily water. Three hundred and forty day visitors draw roughly a tenth of that each. Putting both into one annual total is what makes the figure look harmless, because it divides the peak across eleven months that were never under pressure. Sailings are a proxy for arrivals and arrivals are a proxy for load, and each step of that chain loses the thing the council has to decide.

TAKEAWAY 17W

A visitor is a load rather than a person, and peak load is what the budgets feel.

END OF THE DAY 17W

An aquifer is an account with one deposit, and a licence written above it spends the balance.

DAY 3 OF 15

The thin end of the food web

PLAN CARD 129W

Thursday. Petra Rask, the fisheries observer, is on the island for three days. She wants the trophic arithmetic in the assessment, not in an appendix. Trophic means the feeding levels, and how much energy passes up each one. The west boats land predators. Berhane's transects count what those predators eat. Rask's point is simple. A quota argued in tonnes of fish is really an argument about the bottom of the web. Nobody ever mentions the bottom. Today you work the energy up from the plankton, and say how many species the island can keep. You also say how much a survey of 11 lines can be asked to prove. Argue the top of a web without the bottom, and the quota rests on a number with nothing under it.

BRIEFING 15W

Berhane's transect counts and the west ground's landings describe the same system from two ends.

WHAT YOU DECIDE 20W

Work out how much the reef and the ground can support, and why the top of a web is small.

3.1 Three days a fortnight

HARBOUR OFFICE · A ROOM · VALUE

SCENE 30W

Rask observes three days a fortnight, funded by landing fees. Four of eleven boats landed over quota last season, all of them on days she was not on the island.

GUIDE 49W

Compare the options by what they change about a skipper's decision on an unobserved day, because that is when the over-landing happened. Coverage, penalty size and who does the checking are three different levers. One of them can be paid for out of the fees the island already collects.

ASK 16W

You have four days of observer time this fortnight. Buy what changes what a skipper does.

BEHIND THE BUTTON 140W

Why coverage and penalty trade against each other. A skipper weighs the chance of being caught against what happens if they are. So the same deterrent can be bought two ways. Frequent inspection with a small penalty, or rare inspection with a large one. The island can afford one of those and not the other.

Why landings-book verification is cheap here. Every boat lands at one berth and Calloway is on it. Cross-checking the book against buyers' receipts costs nothing and catches systematic over-landing, though not a single bad trip.

Why peer enforcement works on Vellan and not in general. Eleven boats, all known to each other, all wintering in the same harbour. Published landings turn a private gain into a public one, which is the mechanism that actually held the common's stocking limit for a century before it drifted.

TAKES AS READ

all landings pass across the one berth · carrying capacity, and what happens on the way past it — taken as read

VERDICT 105W

The over-landing happened on days nobody was watching, so the only evidence that changes anything is evidence that reaches those days. Receipts do: every catch is sold, the buyer writes it down, and the comparison against the landings book is retrospective, cheap and total rather than sampled. Observer days at sea are the best evidence about what is caught and they only ever cover the days she is aboard, which are the days nobody offends. A narrower stock estimate moves the quota rather than the compliance. Gear inspection confirms what was already true last year. Both are rigorous, and neither of them reaches the decision.

TAKEAWAY 17W

Deterrence is the chance of being caught times what happens then, and coverage is the affordable half.

3.2 Half the ground, a sixth of the species

COMMON OFFICE · A PERSON · BALLPARK

SCENE 28W

The ferry paper proposes taking eleven hectares of the common for parking and serviced pitches. Pike has a plant list for the common: 96 species over 62 hectares.

GUIDE 55W

Species richness and area are related by a power law, not a straight line: the exponent for islands is about a quarter. Work out what 51 hectares would hold given what 62 holds, using that exponent. Then compare the loss of species with the loss of area, because they are nothing like the same fraction.

ASK 10W

Estimate the species the common would hold at 51 hectares.

BEHIND THE BUTTON 163W

What the exponent means. Because z is about 0.25, richness goes as roughly the fourth root of area. Halving the area does not halve the species; it costs about sixteen per cent of them. Which cuts both ways — a small protected area keeps far more than its share, and a small loss is not harmless either.

Why islands have their own exponent. On a mainland a patch is restocked constantly from next door. Here it is not. Whatever goes locally extinct has to arrive across open water to come back. So island richness sits lower for the same area, and falls further when area is taken.

What the relationship cannot tell you. Which species. The eleven hectares proposed are the flattest and best-drained part of the common. That is where the two orchids in Pike's list are. Nothing else on the island offers the same conditions. A power law is about counts and is silent about the identity of what is lost.

TAKES AS READ

the same species–area relationship holds across the common · species richness, and what an island holds that a mainland does not — taken as read

VERDICT 110W

The ratio of areas is 0.82, and raised to the power 0.25 that is 0.95, so the count falls from 96 to about 91. An eighteen per cent loss of ground costs about five per cent of the species. That is the general shape of the relationship, and it cuts both ways. A small reserve keeps more than its share. A small loss is never proportional. What the arithmetic cannot say is which five, and that is the part Pike will raise. The eleven hectares proposed are the flat, well-drained end of the common. That is where the two orchids are, and nothing else on Vellan offers the same conditions.

TAKEAWAY 20W

Richness goes as about the fourth root of area, so a small area loss costs fewer species than it looks.

3.3 One sample, in July

WATERWORKS · A ROOM · CHOICE

SCENE 32W

Sorley's chloride reading from the borehole in July was 191 milligrams a litre. The previous four years' July readings were 74, 88, 96 and 121. Nobody has sampled in any other month.

GUIDE 60W

Ask three questions of the number before anybody quotes it: what it is about, over what period, and what it is being compared with. A single reading is evidence about the day it was taken. Five readings all taken in the same month are evidence about that month across five years, which is a stronger claim about a different thing.

ASK 13W

What can the five July readings support, put in front of the council?

BEHIND THE BUTTON 161W

Why July is the worst month to sample and the right month to sample. Abstraction is highest and recharge is lowest, so July is when chloride peaks. That makes a July reading unrepresentative of the year. It also makes it exactly the right measurement for deciding whether the borehole is being over-drawn.

What the series can support. Five Julys rising from 74 to 191 is a trend in the peak. The rise is faster than the abstraction has grown. That is the signature of seawater arriving rather than of concentration by evaporation. That is a claim worth making and it is not the same claim as "the island's water is at 191".

How the misquote happens. The figure enters the pack as "chloride at 191 mg/L", loses its month on the way, and comes back as the island's water quality. Sarah Dunmore, the council clerk, has a rule for this: every figure carries what it is about and over what period.

TAKES AS READ

all five readings were taken at the same borehole by the same method · concentration against load, and why a small pipe can matter — taken as read

VERDICT 112W

Five readings from the same month in five years is a series about that month. What it shows is a rising peak. The peak is rising faster than abstraction, and that is the part which points at seawater rather than at concentration. That is the strongest claim the data carries and it is worth making. Stating a single figure as the island's water drops the month and turns a peak into an average. Confirming intrusion is a step further than five numbers can go without a chloride-to-bromide ratio or a second borehole. And seasonal variation is something these readings cannot show at all, because every one of them was taken in July.

TAKEAWAY 19W

A reading is evidence about the conditions it was taken under, and the period is part of the claim.

END OF THE DAY 16W

Each step up a food web keeps about a tenth of what the step below had.

DAY 4 OF 15

Salt in the borehole

BEFORE THE DAY — TRIAL

The far end opens today, and the truck with it

The tip, the hill and the reef station are three hundred metres and more up the island road and nothing up there has been called on this week. From today they are on the list and the truck is signed out for it. Drive the road once so you know which spur is which before Blessing Okafor, who runs the tip, asks you to look at a cell.

PLAN CARD 119W

Friday. The licence has to be lodged today. Sorley has the recharge figure you measured. She also has a fifth July chloride reading, 58 per cent above the first one. Chloride is the salt in a water sample. Berhane has asked the council to cap pumping at the recharge. Calloway says a cap written this week decides the ferry vote before it is taken. She is right that it does. Today you find how long the store takes to turn over. You say what the tip is putting into it, and what a cap would have to say. A store this slow does not tell you it is empty. It tells you years after the decision that emptied it.

BRIEFING 14W

The July chloride is up again and the licence renewal is due on Friday.

WHAT YOU DECIDE 16W

Find out whether the aquifer is being over-drawn, and how long it would take to recover.

4.1 The wall nobody built

REEF STATION · A ROOM · CHOICE

SCENE 33W

Berhane has eleven springs of transect counts and a bathymetric chart on the wall. The reef runs across the mouth of the bay in a broken line, and the harbour sits behind it.

GUIDE 66W

Ask what would have to be built if the thing were not there. The harbour's swell. The beach that the school uses. The ground the west boats fish, and the nursery those fish come from. Those are four separate benefits, and only one of them appears in any island account. The question is which of the four is doing the work the ferry paper silently assumes.

ASK 14W

Which of these is the reef's service the ferry paper is silently relying on?

BEHIND THE BUTTON 159W

What an ecosystem service is. Something a working system does that would otherwise have to be paid for. A reef breaking a swell is a breakwater; a marsh holding a flood is a reservoir; pollination is a workforce. The term exists because these things are invisible in accounts until they stop.

Why the invisibility is the problem rather than the price. Nobody sends the island a bill for the reef, so no line in the ferry paper falls when the reef declines. The cost appears years later, in dredging, in a sea wall, or in the west ground being fished out because its nursery went.

What the transect counts can and cannot support. Eleven springs of cover on eleven lines is a strong record of change and a weak record of cause. It says the reef is thinning; it does not say which of four pressures thinned it, and Berhane will not be moved past that in a meeting.

TAKES AS READ

the reef's position across the bay mouth is what shelters the harbour · an ecosystem as a budget: what comes in, what leaves, what stays — taken as read

VERDICT 108W

The berth is behind the reef line, and a second sailing means more days when the boat has to work in weather. What makes that a service rather than a fact about geography is that the alternative has a price: a breakwater, built and maintained. The paper assumes the shelter without costing it, which is exactly how these things go missing. The other three are not services this reef provides. A reef is not a quarry here. The freshwater lens is held by the rock and the recharge, not by anything offshore. And the quantity of carbon dioxide a coral reef takes up is small, and largely returned.

TAKEAWAY 14W

A service nobody is billed for is invisible in every account until it stops.

4.2 A small pipe, up-catchment

TIP AND SORTING YARD · A PERSON · BALLPARK

SCENE 34W

The tip's leachate collection runs at 2,300 litres a day and tested at 41 milligrams a litre of ammoniacal nitrogen. Okafor's note says the concentration is well below the discharge limit, which is true.

GUIDE 89W

Ammoniacal nitrogen is the form nitrogen takes in leachate, before anything in the ground turns it into nitrate. A concentration says how strong something is. A load says how much of it there is. Multiply the concentration by the flow to get a mass per day. Then mind the units. Milligrams a litre times litres a day gives milligrams a day. That has to come back to kilograms before it goes in a council paper. Then ask which of the two numbers the stream downstream of it responds to.

ASK 8W

Estimate the daily nitrogen load from the leachate.

BEHIND THE BUTTON 163W

Why the limit is written as a concentration and the harm is done by the load. A limit on strength is easy to sample and easy to meet by dilution, which changes the concentration and not the mass. What accumulates in the ground, in the stream and in the borehole is mass, and mass is what a load measures.

Why this pipe matters out of proportion to its size. Two thousand three hundred litres a day is nothing beside the island's water movements, and the tip sits up-catchment of the borehole. A small mass arriving continuously into a store that turns over once a decade accumulates, which is the same residence-time argument in a different currency.

What the comparison should be against. Not the discharge limit, which this passes, but the load the aquifer receives from everything else — the septic tanks, the fertiliser on the common and the sheep. A load only means something beside the other loads into the same store.

TAKES AS READ

the leachate flow is roughly steady through the year · point and non-point sources, and which one a pipe is — taken as read

VERDICT 101W

Forty-one milligrams a litre times 2,300 litres is 94,000 milligrams, which is 0.094 kilograms a day and about 34 kilograms a year. The concentration passes the limit and the load is the number that matters, because what accumulates in an aquifer is mass. This is also why a limit written as a concentration is weak: it can be met by dilution, which changes the strength and not the mass. Thirty-four kilograms a year sounds small. It belongs beside the other loads into the same store. It does not belong beside a discharge limit it was never meant to be compared with.

TAKEAWAY 13W

> Concentration is strength and load is mass; only mass accumulates in a store.

4.3 The pipe and the field

TURBINE YARD · A ROOM · PROTOCOL

SCENE 34W

The stream below the hill runs past the diesel house, the tip spur and eleven hectares of the common. Three samples along it show nitrogen rising, oil rising once after rain, and chloride flat.

GUIDE 58W

A point source arrives from somewhere you can put a finger on and a non-point source arrives from everywhere at once. The tell is not the substance but the pattern: how the reading behaves in dry weather, after rain, and whether it can be turned off. Read each of the three that way before deciding what to regulate.

ASK 10W

Match each observation to what kind of source it indicates.

BEHIND THE BUTTON 143W

Why the distinction decides the response. A point source is licensed, sampled and enforced against — a pipe has an owner. A non-point source is managed by changing practice across a whole catchment, which is slower, cheaper per unit and far harder to make anybody responsible for.

What rainfall does to each. A point discharge is diluted by rain and its concentration falls while its load stays put. A diffuse source is mobilised by rain and both rise together, which is why sampling only in dry weather systematically misses agricultural pollution.

Why the oil is the interesting one. It appears after rain, which is the diffuse signature, and it comes from a hardstanding at the diesel house, which is a definite place. A yard is a point source that behaves like a non-point one, and it is the case that gets argued about.

TAKES AS READ

the three sample points are on the same stream above the tidal reach · a watershed, and what a catchment collects — taken as read

VERDICT 117W

Each pattern is diagnostic. Nitrogen appearing in the first hour after rain and vanishing in dry weather is being washed off fertilised ground. That happens across the whole catchment. It is managed by changing practice rather than by a licence. A concentration steady in every condition has a continuous flow behind it, which is a pipe with an owner. The oil is the case worth arguing. It comes from one yard and only moves when it rains. So it is a point source with a diffuse signature, and both tools apply. And the flat chloride matters most — it rules this stream out of the salt problem, which means the borehole's chloride is arriving through the rock.

TAKEAWAY 13W

The tell is the pattern in dry and wet weather, not the substance.

4.4 Down from what, over how long — Review

HARBOUR OFFICE · A ROOM · CALLBACK · PROTOCOL

SCENE 37W

Petra Rask, the fisheries observer, has four figures pencilled on the back of her sheet and no headings on any of them. Calloway wants all four in the council pack and will not send them up unlabelled.

GUIDE 73W

Four figures and four descriptions of what a figure can be. Ask of each what kind of quantity it is. An amount standing there at one moment. An amount crossing the ground's edge over a stretch of time. Or the gap between two readings of the same amount. The first kind takes a date. The second kind takes a period. The third is not a rate until somebody says how long it took.

ASK 12W

Match each of Rask's figures to what it is a measure of.

BEHIND THE BUTTON 122W

Why the label decides the argument. The same figure supports opposite claims depending on what it is a measure of. Three hundred tonnes standing on the ground is a fishery in trouble; three hundred tonnes arriving each year is a fishery that can be worked. Nothing in the number itself separates those two readings, and the council pack has to.

Why a difference is the awkward one. A stock and a flow each carry their own unit and their own time word, so a reader who wants to check them can. A bare difference carries neither. It is tonnes, like the stock. It happened over a stretch, like the flow. It is the figure most often quoted with the wrong period attached.

TAKES AS READ

the observer's sheet records the west ground alone rather than all island landings

VERDICT 103W

A budget has three kinds of entry and they are not interchangeable. A stock is an amount present at an instant, so it is quoted with a date. A flow crosses the boundary over a stretch, so it is quoted with a period, and inflow and outflow are the two directions of it. A difference between two readings of a stock is neither. It has the stock's unit and the flow's need for a period. It becomes a rate only when the years between the readings are stated. Mislabel one and every comparison drawn from it is against something of a different kind.

TAKEAWAY 24W

A figure is a stock, a flow or a difference, and which of the three it is decides what it can be compared with.

END OF THE DAY 19W

A store with a long residence time hides the mistake that damaged it, and recovers on the same clock.

DAY 5 OF 15

What the west ground can replace

PLAN CARD 130W

Monday. Rask's assessment came off the Friday boat as a range, not a single number. A range is what the council least wants. Josef Mbeki is a skipper, and he has kept to the quota for nine seasons. He can name four boats that have not. He says he will stop keeping to it if the rule stays unenforced. Calloway wants the recommendation today, because the landing fee schedule hangs on it. Today you work out the largest catch the west ground can replace. You also say what setting it at the edge of a range would do. Set a quota at the top of an estimate and you have assumed the estimate was generous. If it was not, the boats take out more than the ground can put back.

BRIEFING 15W

Rask's assessment lands today and the quota has to be recommended before the Friday boat.

WHAT YOU DECIDE 17W

Set a catch the stock can replace, from a stock estimate that has a width to it.

5.1 A quarter of r times K

HARBOUR OFFICE · A ROOM · BALLPARK

SCENE 29W

Rask's assessment puts the west ground's carrying capacity at 1,600 tonnes, give or take 400, with an intrinsic growth rate of 0.42 a year. Last season landed 168 tonnes.

GUIDE 68W

A stock does not grow fastest when it is full. A full stock is at its capacity and barely grows at all; an almost empty one has nothing to grow from. The fastest growth is in the middle. The largest catch that can be taken indefinitely is that growth: a quarter of the growth rate times the capacity. Work it out, then set it beside last season's landings.

ASK 9W

Estimate the maximum sustainable yield of the west ground.

BEHIND THE BUTTON 177W

Why the maximum sits at half. Growth is proportional to how many there are, and to how much room is left. The product of those two peaks when the stock is at half its capacity. That is the whole of the logistic idea. It is why a fishery managed to keep a stock full catches less than one managed to keep it half full.

Why fishing at the maximum is a knife edge. At exactly the peak, any error or bad year pushes the stock below half. Growth is lower there. The same catch is now too big, and the stock keeps falling. Fisheries set targets below the maximum for that reason, and the range on the capacity is what decides how far below.

What the width of the estimate does. Capacity is 1,600 give or take 400, so the sustainable catch is anywhere between 126 and 210 tonnes. Setting the quota at 210 is setting it on the assumption that the generous end of the estimate was right, which is exactly the choice that ends fisheries.

TAKES AS READ

the stock behaves logistically with the growth rate in the assessment · the tragedy of the commons, and the rule that answers it — taken as read

VERDICT 107W

A quarter of 0.42 times 1,600 is 168 tonnes, and last season landed 168. The fishery is being worked at its theoretical maximum. That sounds like success. It is the least stable place a stock can sit. At the peak, one bad year or one over-landing pushes the stock below half its capacity. Growth is slower there, so the same catch becomes too big and the decline continues. The width is the other half of the answer. Capacity is 1,600 plus or minus 400. The sustainable catch is therefore somewhere between 126 and 210 tonnes. So 168 is the middle of a range rather than a number.

TAKEAWAY 18W

The largest replaceable catch is a quarter of r times K, and it sits on a knife edge.

5.2 Eleven years, at what rate

TIP AND SORTING YARD · A PERSON · BALLPARK

SCENE 43W

Blessing Okafor, who runs the tip, weighs everything that arrives. Last year it was 204 tonnes. The figure has risen about 3.5 per cent a year since the sorting yard opened. The cell has eleven years of volume left at last year's rate.

GUIDE 69W

A quantity growing by the same percentage each year does not add, it multiplies, and the useful way to feel that is the time it takes to double. Divide seventy by the annual percentage. Then set the doubling time against the eleven years the hole is said to have. Decide whether "eleven years at last year's rate" is a forecast, or a sentence with a condition hidden in it.

ASK 10W

Estimate the doubling time of the island's annual waste tonnage.

BEHIND THE BUTTON 153W

Where the seventy comes from. It is the natural logarithm of two, multiplied by a hundred. So it works for any steadily compounding quantity: a population, a landfill, a bank balance, an epidemic. Seven per cent doubles in ten years, one per cent in seventy.

Why "at last year's rate" is the load-bearing phrase. Eleven years of volume divided by a constant annual tonnage is arithmetic. The tonnage is not constant, and a quantity that doubles in twenty years fills a hole considerably sooner than a flat one does. The phrase makes the estimate sound careful while removing the growth.

What the island can do about the rate rather than the hole. What leaves on the ferry costs money per tonne and what stays takes volume. Hollis takes compostable material off Okafor's hands and returns it as topsoil, which is the only stream on the island that is genuinely removed from the arithmetic.

TAKES AS READ

the 3.5 per cent annual rise continues at the same rate

VERDICT 105W

Seventy over three and a half is twenty years, and that number is what turns the eleven-year figure into a claim with a condition. A tonnage doubling in twenty years is up about a third within eleven. So the volume consumed is not eleven times last year's arrivals, but appreciably more. The hole closes several years early. The phrase doing the damage is "at last year's rate", which sounds like caution and is the assumption the whole estimate rests on. What the island can actually move is the rate rather than the hole, and the only stream genuinely leaving the arithmetic is what Hollis composts.

TAKEAWAY 16W

A steady percentage growth doubles in seventy over that percentage, and it fills a hole early.

5.3 Many eggs, or few

REEF STATION · A ROOM · CHOICE

SCENE 38W

Two species come off the west ground. One matures at two years and spawns hundreds of thousands of eggs; the other matures at nine and produces a few dozen young a season. The quota treats them as tonnes.

GUIDE 57W

Sort the two by how they reproduce, then ask what each does after a bad year. Many small offspring with no care is one strategy. A few large ones with a long maturation is the other. They respond to fishing in opposite ways. A quota in tonnes is silent about which of the two it is taking.

ASK 15W

What does the difference between the two species mean for a quota written in tonnes?

BEHIND THE BUTTON 152W

What the two strategies are. An r-selected species matures early, breeds prolifically and dies young; a K-selected one matures late, breeds slowly and lives long. Neither is better — they are answers to different conditions, one to an unpredictable environment and the other to a crowded stable one.

Why it decides recovery. Fish out an r-selected stock and it can rebuild in a few seasons if the habitat holds, because the survivors breed fast. Fish out a K-selected one and recovery takes decades, because nothing in its life history is fast. The same tonnage taken from the two is not the same decision.

Why a tonnage quota hides it. A hundred tonnes of a fast-breeding schooling fish and a hundred tonnes of a slow-maturing predator weigh the same and are not the same harvest. This is why assessments are written per species even when the market and the landing fee are not.

TAKES AS READ

both species are taken by the same boats on the same ground · carrying capacity, and what happens on the way past it — taken as read

VERDICT 102W

A hundred tonnes of a fish that matures at two and spawns in hundreds of thousands is a harvest the stock can replace within a few seasons. A hundred tonnes of one that matures at nine and produces dozens is a harvest that removes breeding adults which cannot be replaced for a decade or more. The weight is identical and the decision is not, which is why assessments are written per species. Protecting the abundant one first inverts the argument. Closing the ground entirely is a real option, but the arithmetic does not require it. It requires the quota to be split.

TAKEAWAY 12W

Two stocks of equal weight are not two harvests of equal cost.

5.4 Nine inches over rock — Review

COMMON OFFICE · A ROOM · CALLBACK · PROTOCOL

SCENE 49W

Ewan Hollis has opened a pit in his own two acres of market garden: two feet of dark, heavy, worked soil over the same rock as the common. Pike wants it read against the pit at the top of the common before either acre is promised to the ferry.

GUIDE 72W

The same four questions as the common pit, on ground that answers them differently. How deep is the material a root can work in? What is the texture doing to the water? Where did the organic matter come from? And what is at the bottom. Three of the four are better here than on the common. The fourth is the same on both, and it is the one that sets the ceiling.

ASK 14W

Match each thing the garden pit shows to what it decides about growing there.

BEHIND THE BUTTON 126W

What forty years of importing does to a profile. Seaweed and sorted compost worked in every autumn raises the organic matter, darkens the surface and deepens the worked layer until it meets rock. It builds depth downwards from the top, because there is nowhere else for it to come from on an island. It stops the moment it reaches the rock face.

Why the drainage reverses the usual advantage. On the common a sandy texture over rock drains in twenty minutes and a dry August bites within a fortnight. A heavier, better worked soil holds that water, which carries a crop through the month. The same property leaves it wet in a late spring. That is the reason Hollis plants a fortnight after the mainland does.

TAKES AS READ

the two acres of garden are the only deep soil on Vellan

VERDICT 101W

A profile is read as a set of limits, and on this island three of the four can be bought and one cannot. Depth, organic matter and texture are all products of forty years of importing seaweed and compost. They are why two acres carry vegetables while sixty-two hectares carry sheep. The rock at the base is the term that does not move. It caps the profile. It caps the water held under it. It is why the garden was built upwards instead of dug deeper. Any plan for another acre is a plan to make soil, not to find it.

TAKEAWAY 22W

Depth in an island soil is imported rather than found, and the rock underneath sets a ceiling no amount of importing lifts.

END OF THE DAY 20W

A stock grows fastest at half its capacity, so the biggest sustainable catch is a quarter of r times K.

DAY 6 OF 15

The rule and the money behind it

PLAN CARD 107W

Tuesday. Calloway has the landing fee schedule open beside Rask's list of four boats. Her argument today is the one she is right about. The observer comes over for three days a fortnight. The landing fees pay for those three days. A second sailing would pay for six. Mbeki says he will fish to the limit like everyone else, if the limit stays a suggestion. Berhane objects that the same sailing brings July water demand with it. Today you decide what enforcement the island can actually run, and what pays for it. A quota nobody enforces costs the boats that keep it. It saves nothing at all.

BRIEFING 14W

Four boats landed over quota last season and nothing happened to any of them.

WHAT YOU DECIDE 12W

Decide how the quota is enforced and what pays for the enforcement.

6.1 A tenth of a tenth

REEF STATION · A ROOM · BALLPARK

SCENE 29W

Berhane has a productivity figure for the bay: the plankton fix about 84,000 kilojoules a square metre a year. The west boats land predators three steps up from that.

GUIDE 62W

Work upward one step at a time. Roughly a tenth of the energy held at each level reaches the one above it, because most of it is spent living, lost as heat or never eaten. Three steps up from the producers is a tenth of a tenth of a tenth. Do the arithmetic before deciding whether the west ground's landings are plausible.

ASK 11W

Estimate the energy reaching the third level up from the producers.

BEHIND THE BUTTON 177W

Where the ninety per cent goes. Respiration takes the largest share, because an animal spends most of what it eats staying alive. The rest is lost as heat, excreted, or simply not eaten. The tenth that transfers is an average across systems and runs from about two per cent to twenty.

Why this is the reason food chains are short. Four steps up from a producer leaves a ten-thousandth of the original energy, which cannot support a population of any size. It is also why a predator at the top of a web is rare, slow to mature and slow to recover. And it is why fishing one is a different proposition from fishing a herbivore.

What it means for the quota argument. Landings are measured at the thin end. A change in the plankton or in what grazes it moves the top of the web, with a lag of years. So a stock assessment that reads only landings sees the consequence long after the cause. That is Rask's objection to a quota argued in tonnes alone.

TAKES AS READ

the ten per cent transfer figure applies at each step in this bay · food webs and trophic levels: who eats whom, and how much of it — taken as read

VERDICT 118W

Three transfers at a tenth each leave a thousandth, so 84,000 kilojoules becomes 84. That is the whole reason the top of a web is thin, and it reframes the quota argument. The west boats are harvesting a level that only ever receives a thousandth of the bay's production. So a small proportional change lower down is a large change to them. Using the 0.9 gives 61,000, which is what is lost rather than what transfers. It is a plausible number with the wrong meaning. The tile is on the board because that is exactly how the mistake is made. The lag matters as much as the size: a change at the bottom reaches the landings years later.

TAKEAWAY 14W

Three steps up a web leaves a thousandth of the energy the producers fixed.

6.2 Per tonne, or per cubic metre

TIP AND SORTING YARD · A PERSON · PROTOCOL

SCENE 34W

Okafor has 204 tonnes a year arriving, a cell with eleven years of volume and a ferry that charges by the tonne. Four streams could leave instead of staying: metal, glass, plastic and compostables.

GUIDE 56W

The tip is short of volume, not of mass, and the ferry charges for mass. Rank the four streams by volume saved per pound spent shipping them, not by how much they weigh or how virtuous each feels. One of the four should not be shipped at all, because the island has a use for it.

ASK 11W

Match each stream to what should happen to it, and why.

BEHIND THE BUTTON 156W

Why the two currencies pull apart. Glass is heavy and compacts to almost nothing; plastic is light and takes volume no matter what is done to it. Ship by weight and the plastic is nearly free to remove, while the glass is expensive. That is the opposite of the intuition that heavy things are the problem.

Why compostables are a special case. They are the heaviest stream and the wettest, so shipping them is the worst pound-for-pound choice. Hollis will take them for nothing. He returns them as the only deep topsoil on the island. A stream with a local use is not a waste stream.

What the ferry's own arithmetic adds. Every tonne shipped is a tonne of fuel burnt across the crossing, and the island counts that in its own emissions because the sailing is made for it. The cheapest stream to remove by volume is not automatically the best one to remove overall.

TAKES AS READ

the ferry's charge is per tonne regardless of what the tonne is · carrying capacity, and what happens on the way past it — taken as read

VERDICT 113W

The tip is short of volume and the ferry charges for mass, so the ranking is volume removed per pound of freight. Plastic wins that comparison outright: it is the bulkiest stream and the lightest to send. Glass loses it — heavy to ship and almost free to store once crushed. Compostables are the heaviest, wettest freight and the only stream with a use here, which takes them out of the arithmetic entirely and returns them as topsoil. Scrap is the exception that pays its own passage. The intuition this defeats is that heavy things are the problem; on an island with a hole rather than a weight limit, bulk is the problem.

TAKEAWAY 19W

A tip runs out of volume and a ferry charges for mass, and the two rank the streams differently.

6.3 A July evening, boat in

TURBINE YARD · A ROOM · CHOICE

SCENE 38W

Aled Ferris, the power engineer, has four years of half-hourly demand. The annual average is 88 kilowatts and the highest half hour on record is 310, on a July evening with the ferry alongside and the cable out.

GUIDE 60W

Two different questions get asked of the same data and the answers are in different units. How much energy the island uses in a year is one; how much power the plant has to be able to deliver at once is the other. Work out which of the two sizes the diesel sets, and which one sizes the fuel order.

ASK 13W

Which figure does a second sailing change in the way that costs money?

BEHIND THE BUTTON 156W

What each number decides. Average demand times hours in the year is the energy, which sets the fuel bill and what a turbine could displace. The peak half hour sets the plant: three sets exist so that the largest can fail during the peak and the island stays lit.

Why the peak is where it is. A July evening with the ferry alongside is cooking, showers, the berth's own load and the chandlery all at once, with the cable out. It is not a typical July evening; it is the worst credible combination, and that is exactly what a plant is sized against.

Why a second sailing matters here out of proportion. It does not move the annual energy much: a few more crossings against a whole year. But it adds directly to the one half hour that already sets the plant. A small change to the average can be a large change to the peak.

TAKES AS READ

the four years of half-hourly records cover the same load · renewable and non-renewable, and what the cable is bringing — taken as read

VERDICT 107W

A handful of extra crossings adds very little to a year's energy. It lands directly on the half hour that already sets the plant. That is a July evening with the boat alongside and the cable out. Plant is bought for peaks and fuel is bought for totals, and the two questions have different units: kilowatts against kilowatt-hours. That is why the annual average is the wrong figure to argue the ferry from, and why Ferris keeps returning to a single half hour four years ago. The turbine's capacity factor is a property of the wind rather than of the ferry and does not move at all.

TAKEAWAY 16W

One number buys equipment and the other buys fuel, and they are not the same number.

6.4 What the rain leaves behind — Review

WATERWORKS · A ROOM · CALLBACK · SEQUENCE

SCENE 42W

Sorley has drawn the island's rainfall year on the back of a licence form as four boxes with nothing written on them. Grace Nkemdi, the island nurse, wants it as a poster for the school, and the boxes are in no order.

GUIDE 57W

Four stages, and the rail runs in time. Ask of each what has to have happened already for it to be possible at all. Then ask where in the path the island loses the greater part of what fell. One of the four is the only stage anybody on Vellan meters, and it is not the first.

ASK 14W

Order the four stages by when they happen to one year's rain, earliest first.

BEHIND THE BUTTON 124W

Why the losses come before the store rather than after it. Evaporation, transpiration and runoff all act on water that is still at or near the surface. Once it is past the root zone it is out of reach of the turf and the burns. That is why the fraction reaching the aquifer is settled within days of the rain falling.

What the order buys you on a licence form. Each stage is what is left of the stage before it. So an abstraction limit written against rainfall is written against the largest number in the chain, not the smallest. The licence has to be set at the last stage that is still a natural quantity, and that is the store, not the sky.

TAKES AS READ

the burns on Vellan run to the sea rather than into the aquifer

VERDICT 94W

The chain runs in one direction and each link is smaller than the one above it. Rain arrives on the whole catchment. The turf, the wind and the burns take the greater part of it back before it is anywhere useful. Only what gets past the root zone reaches the store, and only what the pump draws reaches a tap. Putting abstraction before recharge is the common error, and it is the one that matters. It is the shape of a licence written against rainfall instead of against the store the pump actually empties.

TAKEAWAY 28W

Each stage of an island water cycle is what is left of the one before it. A limit is only honest at the stage it is written against.

END OF THE DAY 15W

A rule is only a rule to the extent somebody is funded to check it.

What a kilowatt-hour costs here

PLAN CARD 121W

Wednesday. The cable to the mainland was out for nine days in March. Aled Ferris, the power engineer, has the year's diesel bill. He also has a wind turbine on the hill. It has waited two years for a gearbox. Calloway's ferry paper assumes the turbine comes back. Ferris will not sign that assumption on a brochure figure. He wants a capacity factor measured on Vellan itself. Berhane is on Calloway's side today, for once. Anything that pushes out diesel helps the reef. Today you work out what the island actually uses, in the units a decision can be made in. Argue energy in the wrong unit and the plant comes out half the size it needs, or twice the price.

BRIEFING 16W

The sea cable failed twice in March and the diesel bill for the year is in.

WHAT YOU DECIDE 11W

Put the island's energy use in the units that decide anything.

7.1 Eighty-eight kilowatts, all year

TURBINE YARD · A ROOM · VERIFY

SCENE 35W

Ferris has the island's average demand at 88 kilowatts across the whole year. The diesel invoice covers the nine days the cable was out and he wants the annual energy figure to compare it against.

GUIDE 60W

Energy is a rate multiplied by a time. Take the average power and the hours in a year and multiply. The number that comes out is the whole island's consumption. It is the figure a turbine's output, a fuel bill and a tariff all have to be compared with. None of those can be compared with a demand in kilowatts.

ASK 14W

Predict the island's annual energy from the average demand, then read the meter total.

BEHIND THE BUTTON 149W

Why this conversion is where the confusion lives. A kilowatt is a rate, like a speed; a kilowatt-hour is an amount, like a distance. Most arguments about energy on this island have been two people using the same word for both, and a factor of 8,760 between them.

What the average conceals and why it is still the right number here. It hides the July peak entirely, which is a separate question about plant. For fuel, tariffs and anything a turbine might displace, the total is what matters, and the total is the average times the hours.

Why the cable's nine days show up so clearly. On cable power the island imports; on diesel it burns about a quarter of a litre for each kilowatt-hour. Nine days at 88 kilowatts is around nineteen thousand kilowatt-hours, and the invoice is the proof of the conversion working in the other direction.

TAKES AS READ

the 88 kilowatt average is a genuine annual mean · renewable and non-renewable, and what the cable is bringing — taken as read

VERDICT 113W

Eighty-eight kilowatts held for 8,760 hours is 771 megawatt-hours, and the meter says 764. That is close enough to confirm the average is a real average rather than a convenient one. That total is what everything else is compared against: the fuel bill, the tariff and whatever the turbine might displace. Predicting from the 310 kilowatt peak instead gives 2,700. That is the island's consumption if the worst half hour ever recorded ran all year. The meter would have refuted it by a factor of three. Rate and amount are the same word in ordinary speech and a factor of nearly nine thousand apart here, which is why the reading is worth taking.

TAKEAWAY 13W

Energy is power times time, and a year is 8,760 hours of it.

7.2 Lifting water is most of the bill

WATERWORKS · A ROOM · CHOICE

SCENE 32W

Sorley's borehole pump runs at 11 kilowatts for about nine hours a day. It is the largest single load on the island after the school's heating, and it lifts water 46 metres.

GUIDE 57W

Ask where the electricity goes. Some of it becomes lifted water and the rest becomes heat — in the motor, in the pump and in friction down the rising main. Rank the four places it could be going by how much of the bill each accounts for, then decide which of them is worth spending money on.

ASK 12W

Which part of the pumping bill is worth spending money on first?

BEHIND THE BUTTON 159W

What a pump's efficiency is made of. Motor losses of a few per cent, pump losses of rather more, and pipe friction that rises with the square of the flow. On an old installation the pipe is usually the largest single loss and the cheapest to fix, by slowing the pump and running it longer.

Why the leak is not an efficiency question and matters more. A fifth of what this pump lifts never arrives, which means a fifth of the energy is spent on water that runs into the ground. No improvement in the pump touches that, and it is the largest number in the whole analysis.

Why the lift itself cannot be improved. Forty-six metres of head is where the water is and where the tanks are. No pump can better it. That portion of the bill is physics rather than engineering. The distinction between "loss" and "the work" is the whole of reading an energy account.

TAKES AS READ

the pump runs at roughly constant load when it is running · energy against power: a kilowatt is not a kilowatt-hour — taken as read

VERDICT 113W

A fifth of everything this pump lifts never reaches a tap. So a fifth of the energy is spent on water that goes into the ground. No pump improvement recovers a penny of it. That is the largest recoverable item by a wide margin. The lift itself is not a loss at all: 46 metres of head is the work being paid for, and it cannot be improved by anybody. Motor losses on a modern induction motor are a few per cent. Pipe friction is worth attacking after the leak, not before. Slowing the flow to reduce friction means running longer, and that only helps once the water arriving is water being used.

TAKEAWAY 20W

Some of an energy bill is the work and the rest is loss, and only the loss can be recovered.

7.3 A threshold, not a calendar

COMMON OFFICE · A PERSON · CHOICE

SCENE 33W

Hollis has aphids in two of four polytunnels. The neighbouring holding sprays on the first of the month regardless. His tunnels need treating twice a season; theirs need it five times and rising.

GUIDE 81W

What Hollis does has a name: integrated pest management, which means every tool the crop allows and the spray only when a count says so. Compare the two regimes by what they do to the pest population over years, not by what they cost this month. Ask which one selects for resistance, and how a threshold is set. "Treat when you see one" and "treat when it will cost you more than the spray does" are different rules, with different outcomes.

ASK 12W

Why does the neighbouring holding need five sprays where Hollis needs two?

BEHIND THE BUTTON 134W

How calendar spraying breeds the problem. Every application kills the susceptible individuals and leaves the resistant ones to breed, so the population's tolerance rises with each treatment. Five sprays a season rising is what that looks like from inside: the same chemical, doing less each year.

What a threshold is. The pest density at which the damage exceeds the cost of treating. Below it, spraying costs more than it saves and still selects for resistance. Above it, treating pays. Setting it requires counting, which is the part growers skip.

What else integrated pest management uses. Resistant varieties, timing of sowing, encouraging predators, and physical barriers — with the spray as the last resort rather than the schedule. Hollis's two treatments a season are the residue after the other four tools have done their work.

TAKES AS READ

both holdings grow comparable crops under comparable cover · agricultural methods, irrigation and the salt they leave behind — taken as read

VERDICT 97W

Each application kills the susceptible aphids and leaves the resistant ones to breed, so a fixed schedule raises the population's tolerance season by season. Five treatments and rising is the signature: the same chemical achieving less each year. Hollis's two are what is left after resistant varieties, sowing dates, predators and barriers have done the rest. The spray is kept as a last resort, against a counted threshold. Temperature and crop susceptibility would show as a difference from the first season, not a worsening trend. A stronger chemical would accelerate the same selection rather than avoid it.

TAKEAWAY 13W

Treating on a calendar selects for resistance; treating on a threshold does not.

7.4 The wall nobody built — Review

REEF STATION · A ROOM · CALLBACK · CHOICE

SCENE 36W

Berhane has a second chart out: four hectares of seagrass inside the bay, between the pier and the burn mouth. A quote has come across from the mainland for dredging a deeper channel straight through it.

GUIDE 66W

Ask what the island would have to start doing for itself if the bed were gone. Three of the four candidates describe work that something else on the island already does, or that no marine plant does at this scale. One of them is work the harbour currently gets for nothing and would have to buy every year. The quote prices the dredging and not that.

ASK 18W

Which service would the island have to start paying for if the channel is cut through the bed?

BEHIND THE BUTTON 115W

Why a one-off quote hides an annual cost. Cutting the channel is a single payment and the service it removes was a payment avoided every year. A cost that recurs is worth far more than a cost that happens once. The paper in front of the council shows only the second of them. Only the second has an invoice attached.

How to tell a service from a fact about the place. The test is substitution. If the thing stopped, would somebody have to build or buy a replacement? A reef breaking a swell substitutes for a breakwater. A view from the pier substitutes for nothing, and is not counted here whatever else it is worth.

TAKES AS READ

the berth is dredged only when it has silted enough to shoal

VERDICT 101W

Seagrass roots bind the soft sediment of an inner bay, and the leaves slow the water over it. So the fines settle in the bed instead of travelling on. The deepest hole in the bay is the berth. Cut the channel and that stops, and the island buys a dredger visit on a cycle for as long as the harbour is used. That is the substitution test passing. The other three fail it. The garden's seaweed comes off the strandline. No marine plant cleans a diesel plume. And the shoreline seepage is the island's own lens coming out through the rock.

TAKEAWAY 20W

The test of a service is substitution: what the island would have to build or buy if the thing stopped.

END OF THE DAY 15W

Power is the rate and energy is the amount, and the two decide different things.

DAY 8 OF 15

What leaks out of the hole

BEFORE THE DAY — FOLLOW

Follow Pike round the stocking count

Dan Pike, the common grazing officer, counts stock on the common twice a year and he does not stop walking to explain it. Stay with him: the count is the only evidence behind the stocking limit, and the argument at the spring meeting is always about how it was taken rather than what it said.

PLAN CARD 138W

Thursday. Grace Nkemdi, the island nurse, has asked for the school tap result to be read out at the council. She does not want it filed. Okafor's leachate is past its concentration limit. Leachate is the liquid that drains out of the tip. It is adding 34 kilograms of nitrogen a year to a store that turns over once a decade. Sarah Dunmore, the council clerk, wants every figure in the pack to name what it is about. Today you convert between the units the licence, the laboratory and the clinic each use. You then say which of them the school tap answers to. Two of these numbers are the same measurement in different clothes, and one of them is not. Mix them up and the school tap is judged against a limit that was never about it.

BRIEFING 17W

The stream below the tip has nitrogen, and the school tap is sampled separately for a reason.

WHAT YOU DECIDE 15W

Follow the tip's leachate into the water it reaches and say what the numbers mean.

8.1 Milligrams, or parts per million

TIP AND SORTING YARD · A ROOM · BALLPARK

SCENE 37W

The laboratory reports the stream below the tip at 8.4 milligrams a litre of nitrate. The licence is written in parts per million and the mainland guidance in micrograms a litre. All three describe the same water.

GUIDE 52W

A litre of fresh water weighs about a kilogram, so a milligram dissolved in it is one part in a million by mass. That makes two of these three units the same number. Convert the laboratory's figure into the licence's units, and into the guidance's, and check which pair are actually identical.

ASK 8W

Estimate the stream's nitrate in micrograms a litre.

BEHIND THE BUTTON 167W

Why the approximation holds here and not everywhere. Fresh water at ordinary temperatures is close enough to a kilogram a litre that the conversion is exact for any purpose a council paper has. In seawater, or in a brine, the density has moved and so has the conversion — which matters on an island where some samples are brackish.

Why three units exist for one quantity. The licence was drafted from a national template in parts per million. The laboratory reports in milligrams a litre, because that is what its instruments produce. Health guidance uses micrograms, because the numbers of interest are small. Nobody is being obscure; the units accumulated.

What the conversion cannot fix. None of the three is a load, and none is a dose. The same 8.4 milligrams a litre is a different mass per day depending on the flow and a different dose depending on who drinks it and how much. Getting the units right is the first step and not the answer.

TAKES AS READ

the sample is fresh water rather than brackish

VERDICT 103W

A milligram is a thousand micrograms, so the laboratory's 8.4 becomes 8,400 — the same water, reported in a unit chosen because health numbers are small. The parts-per-million figure is the one worth noticing. In fresh water a milligram a litre is a milligram in a kilogram, which is one part in a million. So 8.4 mg/L and 8.4 ppm are not merely comparable but identical. The density tile is what that rests on, and it stops being one in brackish water, which is half the samples on this island. What none of the three units gives is a load or a dose.

TAKEAWAY 14W

Milligrams a litre and parts per million are the same number in fresh water.

8.2 **Nutrients in, oxygen out**

REEF STATION · A PERSON · SEQUENCE

SCENE 38W

Berhane has dissolved oxygen readings from the inner bay: 8.1 milligrams a litre in March, 4.2 in August, and 2.9 at dawn on the third of August. The nitrate arriving from the stream peaks in the same month.

GUIDE 71W

Berhane calls this eutrophication, which is the course's word for nutrients in and oxygen out. Follow the nitrogen through four steps rather than two. It feeds something, that something dies, the dying is what uses the oxygen, and the oxygen is what everything else needed. Then look at why the worst reading is at dawn, because the timing is the evidence that the mechanism is the one you think it is.

ASK 9W

Put the four steps in the order they happen.

BEHIND THE BUTTON 148W

Why dawn is the diagnostic hour. Algae photosynthesise in daylight and respire all the time, so oxygen peaks in late afternoon and bottoms out just before sunrise. A dawn minimum well below the afternoon reading is the signature of a large algal population rather than of a single discharge.

Why warm water makes it worse twice over. Warm water holds less dissolved oxygen to begin with, and it speeds up both algal growth and the decomposition that consumes the oxygen. August is the month the nutrients arrive, the water is warmest and the oxygen ceiling is lowest.

What suffers first. Not the algae and not the bacteria. Fish and the invertebrates on the reef leave or die. The ones that cannot move are the ones counted on Berhane's transects. That is how a nutrient problem in a stream becomes a line on a reef survey two months later.

TAKES AS READ

the oxygen readings were taken at the same depth in the inner bay · concentration against load, and why a small pipe can matter — taken as read · the carbon and nitrogen cycles, and where a farm interrupts them — taken as read

VERDICT 101W

The chain has a step in it that is easy to leave out: the algae themselves do not take the oxygen, the bacteria decomposing them do. That is why the damage arrives after the bloom rather than during it, and why the water can look clear on the day the oxygen is worst. The dawn reading of 2.9 is the confirmation — oxygen is lowest just before sunrise because nothing has been photosynthesising all night while everything has been respiring. And August stacks three things: the nutrients arrive, the water is warmest, and warm water holds less oxygen to start with.

TAKEAWAY 13W

> Nutrients feed algae, decomposition spends the oxygen, and what needed the oxygen leaves.

8.3 Eleven kilos

HARBOUR OFFICE · A ROOM · CHOICE

SCENE 29W

Nkemdi wants the school tap read out at the council. The distribution main tested at 8.4 milligrams a litre; the school tap, sampled after the weekend, tested at 11.2.

GUIDE 56W

A sample is a claim about the water in one place at one time. Ask what makes the school tap different from the main — where it sits on the network, when the sample was taken, and who drinks from it. Two of those three change the number and the third changes what the number means.

ASK 9W

What is the school tap result a result about?

BEHIND THE BUTTON 161W

Why the end of a main reads higher. Water sits longest there, and a weekend with the school closed means it sat for two days. Anything that accumulates has had longer to accumulate. A Monday sample at the far end is therefore systematically the worst reading on the network. It is a real reading rather than an artefact.

Why Nkemdi asked for it separately. Concentration is the same water for everybody and dose is not. The youngest child at that tap weighs eleven kilograms and drinks about a litre a day. An adult drinking two litres takes roughly a sixth of that dose per kilogram. The concentration is identical.

What the council can be told. That the school tap is the highest reading on the network. That the Monday timing partly explains it. And that the dose it implies for the smallest children is what the guidance is written for. All three are defensible; any one of them alone is misleading.

TAKES AS READ

both samples were taken by the same method in the same week · concentration against load, and why a small pipe can matter — taken as read

VERDICT 116W

Two things are true at once and the honest sentence contains both. The school tap is at the end of the main. It was sampled after a weekend of standing water. That is why it reads highest, and the reading is real rather than an error. And it is the tap where an eleven-kilogram child drinks, so the same concentration is about six times the dose per kilogram it is for an adult. Quoting it as the island's water quality takes the worst point on the network as the average. Calling it contamination in the building asserts a source nothing has looked for. Calling it a sampling error dismisses the highest genuine exposure the network has.

TAKEAWAY 12W

A concentration is the same for everybody, and a dose is not.

END OF THE DAY 12W

Concentration, load and dose are three different questions about the same water.

What is put on the ground

PLAN CARD 109W

Friday, and two requests arrived on the same boat. Ewan Hollis, the island's market gardener, wants the whole compostable stream released to him. That would take 40 tonnes a year out of Okafor's hole. The neighbouring holding wants its spray allowance raised. That puts more of the same chemical on ground that drains past the borehole. Nkemdi has worked out the dose for the school tap. She wants it in the council pack. Today you decide what goes on the ground, and what a dose actually is. A dose is not simply an amount. The same milligram is a different quantity depending on whose body it ends up in.

BRIEFING 14W

Hollis wants more compost released and the neighbouring holding wants the spray limit raised.

WHAT YOU DECIDE 14W

Weigh the inputs on the common and the garden against what they do downstream.

9.1 The same water, a different dose

WATERWORKS · A PERSON · BALLPARK

SCENE 32W

The school tap reads 11.2 milligrams a litre. Nkemdi's youngest pupil weighs 11 kilograms and drinks about a litre a day; an adult on the same tap drinks two and weighs seventy.

GUIDE 58W

A dose is not a concentration. Multiply the concentration by how much is drunk to get the amount taken in, then divide by the mass of the body taking it. Do it for the child, then set it against the same arithmetic for an adult. The ratio is the whole reason guidance is written the way it is.

ASK 8W

Estimate the daily dose for the eleven-kilogram child.

BEHIND THE BUTTON 151W

Why the guidance is per kilogram. A tolerable intake is a property of a body's ability to process something, which scales roughly with its mass. A figure quoted per person is a figure written for an average adult, and the smallest person drinking the water is the one it fails.

Why intake matters as much as mass. Children drink more per kilogram than adults do — a litre in eleven kilos against two in seventy — so the two factors compound rather than cancel. That is why the ratio between the two doses is far larger than the ratio of the body masses.

What a dose-response curve adds. Whether the dose matters depends on where it sits on a curve, and that curve is usually not a straight line. Below a threshold, little or nothing. Past it, rising harm. The dose is the input to that judgement rather than the judgement.

TAKES AS READ

the consumption figures are typical daily intakes

VERDICT 105W

Eleven point two milligrams in a litre, one litre drunk, divided by eleven kilograms, is about one milligram per kilogram a day. The adult on the same tap takes 22.4 milligrams into seventy kilograms, which is 0.32 — so the identical water is three times the dose for the child. The two factors compound instead of cancelling, because children drink more per kilogram as well as weighing less. That is why guidance is written per kilogram and why a figure quoted per person is written for an adult. Using the adult mass with the child's intake mixes two people and produces a dose nobody receives.

TAKEAWAY 12W

A dose is intake times concentration, divided by the body taking it.

9.2 Forty tonnes, twice

COMMON OFFICE · A ROOM · TRIGGER

SCENE 33W

Hollis wants the whole compostable stream: about forty tonnes a year, currently going into the hole. He has two acres of built soil and four polytunnels, and no fertiliser budget to speak of.

GUIDE 61W

Count what the transfer does at both ends. Forty tonnes is volume out of a tip with eleven years left, and it is organic matter into the only deep soil on the island. Then find the cost, because there is one. The same material carries salt and whatever was sprayed on it. It arrives on ground that drains toward the borehole.

ASK 17W

At what compost conductivity do you stop the release, given the ground takes six weeks to flush?

BEHIND THE BUTTON 148W

What compost does to a thin soil. It raises organic matter, which raises how much water the soil holds and how much nutrient it can retain against leaching. On a sandy loam over rock that is the only lever available: nothing else deepens what is there.

Why the salt is worth checking. Island compostables include seaweed and kitchen waste from a place where everything is salted. Repeated application of a salty amendment to free-draining ground raises soil salinity, until germination suffers. The soil is not flushed by irrigation here — it is flushed by rain, in winter, if at all.

What the tip gains and does not. Forty tonnes out of the arrivals slows the hole's filling and reduces the methane the cell generates, because compostables are what generate it. It does not reduce the tonnage the island produces, which is the number Okafor is actually judged on.

TAKES AS READ

the compostable stream is roughly forty tonnes a year as weighed · soil: horizons, texture and what a thin soil cannot do — taken as read

VERDICT 133W

The transfer is good at both ends. Volume comes out of a hole with eleven years left. Organic matter goes into the only deep soil on the island. So the rule has to catch the one thing that makes it go wrong. Island compostables carry seaweed and heavily salted kitchen waste, and on free-draining ground over rock there is nothing to flush salt through but winter rain. Two ways of getting the line wrong cost the same. Set it near 4, where germination measurably suffers. Then it is not crossed until the week-13 batch, with two weeks of rain left. That is a correct rule for an action that needed six. Set it near 1.5 and it fires on ordinary compost, and a rule that fires on normal is a rule nobody keeps.

TAKEAWAY 17W

> Compost is the only lever on a thin soil, and it carries whatever was in the material.

9.3 What the hole gets back

TIP AND SORTING YARD · A ROOM · ALLOCATE

SCENE 33W

Okafor has 204 tonnes arriving, eleven years of volume, and four things she could do with a fixed budget of eleven thousand pounds this year. Every option removes volume at a different price.

GUIDE 61W

Everything here removes volume from the hole and the budget cannot buy all of it. Work out what each pound buys in cubic metres, spend the budget, and watch what happens to the year the cell closes. Two of these options also change what the tip emits, which is worth knowing before choosing between two that look equally good on volume.

ASK 18W

Eleven thousand pounds. Build the year's plan, and know what it stops the tip being able to do.

BEHIND THE BUTTON 138W

Why volume rather than tonnage is the currency. The constraint is a hole. Compaction buys volume without removing any mass at all, which makes it look like cheating and is the cheapest cubic metre available until the compactor is running full.

What diverting compostables does twice. It removes the wettest, heaviest stream from the hole and removes the material that generates most of the methane, because methane comes from organic matter decomposing without oxygen. One action, two benefits, and only one of them shows up in a volume calculation.

Why shipping is priced per tonne and bought per cubic metre. The ferry charges weight and the island needs volume, so the streams worth shipping are the light bulky ones. Spending the whole budget on freight buys fewer cubic metres than spending part of it on a compactor service.

TAKES AS READ

the budget is for this year only and does not roll over · carrying capacity, and what happens on the way past it — taken as read

VERDICT 110W

Volume per pound is the ranking, and it does not follow weight or virtue. Diverting compostables is the cheapest large block and takes most of the methane source with it, which no volume figure shows. Compaction is next: it removes no mass at all and recovers more cubic metres than shipping plastic for a third less money. Scrap is nearly free and pays part of itself back. Glass is last by a wide margin — thirty tonnes of freight for sixty cubic metres, because glass crushes to almost nothing and should stay. Spending the whole budget on freight is the intuitive answer and buys roughly a third of the volume.

TAKEAWAY 15W

> The constraint is volume, and the cheapest cubic metre is not always the obvious one.

END OF THE DAY 16W

Dose is exposure divided by the body taking it, which is why children are the limit.

A warmer, sourer sea

PLAN CARD 121W

Monday. Berhane has this spring's transects on the table, beside the other ten. Coral cover is down a further 4 per cent. The August sea temperature was the highest in the series. Calloway's position today is that the reef's decline is global. Nothing the island does about ferries changes it, she says. That is partly true, and it is being used to end an argument. Dunmore wants the pack to state exactly what 11 springs of counting support. Today you read the record for its trend. You separate what the island controls from what it does not, and say what the reef is running out of. A record this good is easy to overstate. Overstating it is how it gets dismissed.

BRIEFING 11W

Berhane's spring count is in and the cover is down again.

WHAT YOU DECIDE 13W

Read eleven springs of reef data for what it can and cannot say.

10.1 Two degrees, in August

REEF STATION · A ROOM · SWEEP

SCENE 49W

August sea temperature in the bay has risen by nearly two degrees across the eleven springs of record. Cover on the transects has fallen from 41 per cent to 26. The station has a flow tank that will hold a reef fragment at a set temperature for six weeks.

GUIDE 75W

A species lives inside a range rather than at a point, and the interesting part of a range is its edge. The tank holds a fragment at whatever August temperature you set and reports the cover it still has after six weeks. Move the control across the range and watch where the readout stops sloping and starts falling away. Nothing marks it. Two degrees is small on a thermometer and not small at an edge.

ASK 11W

At what August temperature does the fragment stop holding its cover?

BEHIND THE BUTTON 165W

What a tolerance range is. Every population has conditions in which it thrives, wider conditions in which it survives with reduced growth and reproduction, and limits beyond which it dies. Moving the average two degrees moves the whole distribution, so what was the rare bad week becomes the normal August.

Why the two stresses compound. Warmer water expels the symbiotic algae corals depend on for most of their energy, and a more acidic sea makes the skeleton more expensive to build. An animal short of energy and facing a higher construction cost is not facing two problems; it is facing one, twice.

What the island controls and what it does not. Sea temperature and acidity are set far away. Nutrient load from the stream. Sediment from the works. Anchor damage in the bay. Fishing pressure on the grazers that keep algae off the coral. All four are local, and all of them decide how much stress the reef meets on top of the global part.

TAKES AS READ

the eleven transects have been read by the same method each spring · greenhouse gases, and comparing them on one scale — taken as read · ecological tolerance, and the range a population can actually live in — taken as read

VERDICT 127W

A tolerance range is not a switch and it is not a straight line either. Between 15 and 17.5 degrees the fragment holds nearly all its cover. The curve then falls away steeply. So the animal is not gradually worse across its range. It is fine until it is not. The two-degree rise the bay has actually seen lands exactly on that shoulder. That is why cover has gone from 41 per cent to 26, rather than from 41 to 38. It also settles what the island can do. At the shoulder there is nothing spare. Not for a nutrient

> pulse, a sediment plume, an anchor scar or a summer with no grazers. Every one of those is decided on Vellan, even though the temperature is not.
>
> **TAKEAWAY** 16W
>
> A warmed animal at the edge of its range meets every other stress with nothing spare.

10.2 What eleven springs can say

TURBINE YARD · A ROOM · PROTOCOL

SCENE 36W

The pack has three draft sentences about the reef in front of it. Ferris wants the one that will survive being read out by somebody who disagrees with it; Dunmore wants the one that is true.

GUIDE 52W

Take each draft sentence and ask what evidence it would take to support it. Then check whether eleven springs of cover and temperature on one bay is that evidence. A trend, a correlation and a cause need three different amounts of data, and only one of the three is in the room.

ASK 12W

Match each draft sentence to what it would take to support it.

BEHIND THE BUTTON 135W

What a trend needs. Repeated measurement by a consistent method over a period long relative to the variation. Eleven springs on eleven fixed transects is exactly that, and it is why this record is the strongest thing on the island.

What attribution needs beyond a trend. Either an experiment, which is impossible here, or enough independent lines of evidence to rule out the alternatives. Other reefs as controls. A bleaching record tied to specific warm weeks. A nutrient series. A sediment series. Some of that exists elsewhere; almost none of it exists for Vellan.

Why understating is also a failure. "Cover has changed" is defensible and useless. The pack has to carry the strongest claim the evidence supports. A paper that says less than it knows invites a decision the evidence would not have allowed.

TAKES AS READ

the transect method has not changed across the eleven springs · concentration against load, and why a small pipe can matter — taken as read

VERDICT 117W

The record is strong and it is strong about one thing: a fall in cover, measured the same way, over long enough to be a trend. Putting the temperature series beside it gives a correlation, which is a real and weaker claim. Attribution is a different order of evidence — control sites, bleaching tied to specific warm weeks, the alternatives eliminated — and none of that exists here. The nutrient sentence is the one to strike outright, because nobody has collected a nutrient series for the bay. The pack should carry the two claims the data supports, stated at full strength. A paper that says less than it knows invites a decision the evidence would have prevented.

TAKEAWAY 13W

A trend, a correlation and a cause need three different amounts of evidence.

10.3 What the grazers were doing

HARBOUR OFFICE · A PERSON · CHOICE

SCENE 32W

The reef's algal cover has risen as coral cover has fallen. Rask's landings show the grazing species — taken as a low-value bycatch — down by a third over the same period.

GUIDE 52W

Follow what eats what. Something is keeping algae off the coral. The landings of that something have fallen a third. So the question is whether the algae rose because the coral weakened, or because its grazers went. Both mechanisms are consistent with the same two graphs, and they call for different actions.

ASK 12W

What does the fall in grazer landings imply about the reef's recovery?

BEHIND THE BUTTON 130W

What grazers do on a reef. They crop algae that would otherwise overgrow coral and stop new coral settling. Remove them and algae wins the space even where conditions would let coral recover, which turns a temporary stress into a permanent change of state.

Why a low-value bycatch is the dangerous one. Nobody manages a species nobody sells. The grazers here are taken incidentally, not counted carefully and not in the quota, so a third of them can go without any part of the system noticing.

Why this is the island's lever. Sea temperature is set elsewhere. Whether the reef keeps its grazers is decided at this berth, by what the quota counts. It is the one part of the reef's problem that a council on Vellan can vote on.

TAKES AS READ

the grazing species landings are recorded even as bycatch · an ecosystem as a budget: what comes in, what leaves, what stays — taken as read

VERDICT 99W

Grazers crop the algae that would otherwise overgrow coral and block new coral from settling. Take a third of them out and algae keeps the space in the good years too. A temperature stress that would have been temporary becomes a permanent change of state. That is why this stop is on the harbour's list rather than the reef's. The temperature is set far away. The grazers are decided at this berth, by what the quota counts. A low-value bycatch is exactly the case nobody manages, which is what makes a third of them disappearing possible without anybody objecting.

TAKEAWAY 13W

Removing the grazers turns a temporary stress into a permanent change of state.

END OF THE DAY 15W

A long local record can show a trend clearly and still not settle a cause.

DAY 11 OF 15

What the diesel actually buys

PLAN CARD 114W

Tuesday. Ferris has the fuel delivery notes for the whole year. They add up to 78,000 litres, burnt in three generator sets. Those sets made most of the island's electricity while the cable was out, and some of it when it was not. Calloway needs the number for the ferry paper. Berhane wants the emissions from it stated in the same place, not in an annex. Dunmore will not accept a figure without its period attached. Today you work out how much of that fuel became electricity, and what the rest became. Most of a litre of diesel does not become electricity. Where the rest goes decides whether the turbine is worth its gearbox.

BRIEFING 15W

The year's fuel figure has to go in the pack next to the turbine proposal.

WHAT YOU DECIDE 15W

Account for the fuel: what becomes electricity, what becomes heat, and what leaves the stack.

11.1 A third, at best

TURBINE YARD · A ROOM · BALANCE

SCENE 29W

Seventy-eight thousand litres of diesel went into the sets this year. Each litre holds about 10.7 kilowatt-hours of energy, and the sets generated 271,000 kilowatt-hours of electricity between them.

GUIDE 48W

Work out the energy that went in, then compare it with the electricity that came out. The ratio is the efficiency. Then account for the missing part: it has not disappeared, and on this island there is a use for some of it that nobody has taken up.

ASK 14W

Read what you need and report the energy that left the sets as heat.

BEHIND THE BUTTON 172W

Why a third is normal. A diesel generator turns about a third of the fuel's chemical energy into electricity at full load, less at part load. The rest leaves as exhaust heat and jacket cooling. This is not a fault in the machine — it is what a heat engine does, and the ceiling is set by thermodynamics rather than maintenance.

Why part load makes it worse. A set running at a quarter of its rating burns a disproportionate amount of fuel for what it delivers. Vellan's demand varies by a factor of three across a day. Running one set hard is better than three sets lightly, which is a scheduling decision rather than a capital one.

What the heat could do. Two thirds of 835,000 kilowatt-hours is a very large amount of low-grade heat. It leaves the diesel house through a radiator forty metres from a school that burns oil for its heating. Nobody has piped it, and no efficiency figure for the generator will ever show that as a loss.

TAKES AS READ

the fuel's energy content is about 10.7 kilowatt-hours a litre

VERDICT 127W

Seventy-eight thousand litres at 10.7 kilowatt-hours each is 835 megawatt-hours in. Of that, 271 reached the board. A third is what a diesel generator does when it is working properly. The term that does not announce itself is the auxiliaries. Pumps, fans and jacket heaters are drawn by the sets before the board meter. So they are never billed and never appear. Forty-two megawatt-hours of work the plant does for itself. What is left, 522 megawatt-hours, is low-grade heat. It leaves through a radiator forty metres from a school heated with oil. No efficiency figure for the generator will ever record that as a loss. The jacket cooling is part of that heat rather than a separate stream, and counting it again double-counts the largest number here.

TAKEAWAY 20W

A third of a fuel becomes electricity and the rest leaves as heat, which is a resource or a loss.

11.2 Twenty-eight to one

TIP AND SORTING YARD · A PERSON · BALLPARK

SCENE 31W

The cell vents methane from decomposing organic matter. Okafor's estimate is 14 tonnes a year. The diesel house emits about 205 tonnes of carbon dioxide. The pack lists only the second.

GUIDE 59W

Two gases cannot be added until they are in the same unit. Each has a global warming potential — how much heat a kilogram of it traps compared with a kilogram of carbon dioxide, over a stated period. Convert the methane and then compare, and note which period the conversion factor belongs to, because the answer changes with it.

ASK 9W

Estimate the tip's methane emissions as carbon dioxide equivalent.

BEHIND THE BUTTON 161W

Why methane's factor is so large and so slippery. It traps far more heat per kilogram than carbon dioxide, and it breaks down in about a decade. So its potential depends entirely on the period chosen. Roughly 28 over a hundred years, and around 80 over twenty. Both numbers are correct and they describe different questions.

Why a landfill emits it at all. Organic matter decomposing without oxygen produces methane rather than carbon dioxide. That is why diverting the compostable stream is an emissions action as much as a volume action. It is also why a flare is worth so much. A flare burns methane to carbon dioxide, and it is worth that despite producing a greenhouse gas itself.

Why the pack lists only the diesel. Fuel arrives on an invoice and gas leaves through a vent nobody meters. What is easy to count gets counted, and an island reporting only its fuel emissions has left out something comparable in size.

TAKES AS READ

the methane estimate is for a year of the cell's current arrivals · the carbon and nitrogen cycles, and where a farm interrupts them — taken as read

VERDICT 135W

The diesel is counted from the delivery note rather than the chimney: 78,000 litres at 2.63 kilograms a litre is 205 tonnes. Fourteen tonnes of methane at a hundred-year potential of 28 is 392 tonnes of carbon dioxide equivalent. The island's largest single greenhouse source is a vent on a hole. It is not in the pack, because fuel arrives on an invoice and gas does not. The period is doing real work in that sentence: at the twenty-year potential of 80 the same 14 tonnes is 1,120, which is five times the diesel house. Both figures are correct and they answer different questions, which is why the pack has to say which one it used. Adding the raw tonnages gives 219 and compares kilograms of two gases as though they were the same substance.

TAKEAWAY 16W

Two gases compare only once both are multiplied to a common unit, over a stated period.

11.3 Fixed, spread, and lost

COMMON OFFICE · A ROOM · PROTOCOL

SCENE 27W

Pike has three nitrogen inputs to the common: clover in the sward, imported fertiliser, and dung from 512 sheep. Hollis adds compost. The stream below carries nitrate.

GUIDE 60W

Sort the inputs by where the nitrogen came from before it arrived here. Some was taken out of the air on the island itself. Some was manufactured elsewhere and shipped in. And some is nitrogen already on the island, being moved around. Only one of the three adds to the island's total, and that is the one the stream reflects.

ASK 12W

Match each input to what it does to the island's nitrogen budget.

BEHIND THE BUTTON 156W

What fixation is. Bacteria in the nodules of clover and other legumes convert nitrogen gas from the air into a form plants can use. It is a genuine input to the island and it arrives slowly, in place, and in proportion to how much clover is in the sward.

Why dung is not an input. The sheep ate the grass here. Their dung returns nitrogen that was already on the common to the common, moved from where it was eaten to where the animal stood. It matters enormously for distribution and not at all for the total.

Why imported fertiliser is the one the stream sees. It is nitrogen manufactured elsewhere, landed at the berth and spread in a few weeks. Applied faster than the sward can take it up, the surplus leaches. Free-draining ground over rock leaches readily. That is the connection between a bag on the pier and a nitrate reading in the bay.

TAKES AS READ

the sheep graze the common rather than being fed imported feed · an ecosystem as a budget: what comes in, what leaves, what stays — taken as read

VERDICT 109W

Two of these four add nitrogen to the island and two redistribute what is already on it. Clover fixes it out of the air, slowly and in place. Bagged fertiliser is manufactured elsewhere, landed at the berth and spread over a few weeks. That is why it is the input the stream reflects. It is applied faster than the sward can take up, so the surplus leaches through free-draining ground over rock. Dung and compost feel like additions and are not — the sheep ate grass grown here and the kitchen waste was island food. Getting that distinction right is what stops the response being aimed at the sheep.

TAKEAWAY 14W

Only what arrives from outside adds to the island's total; the rest is redistribution.

END OF THE DAY 16W

Efficiency says what fraction of the fuel did the job, and the rest left as heat.

DAY 12 OF 15

The gearbox and the nine years

PLAN CARD 118W

Wednesday. The gearbox quote came in on the Tuesday boat: £41,000, and 11 weeks to fit. Calloway's ferry paper already assumes the turbine is turning. A diesel island is a hard thing to sell to visitors. Ferris will not sign that assumption without Vellan's own capacity factor. He is right to ask for it. Berhane wants the emissions saving stated as well. Today you work out what the machine really produces here. You also work out what the gearbox pays back, and how long that takes. A nameplate rating is a number for a brochure, not what the hill delivers. If the payback runs to nine years, being wrong about it costs the island nine years of diesel.

BRIEFING 14W

The gearbox quote is in and the ferry paper assumes the turbine is running.

WHAT YOU DECIDE 11W

Cost the turbine honestly, on Vellan's wind rather than a brochure's.

12.1 Rated, and actual

TURBINE YARD · A ROOM · BALLPARK

SCENE 28W

The turbine is rated at 250 kilowatts. In its last full year before the gearbox failed it generated 613,000 kilowatt-hours. The brochure quotes an annual output of 900,000.

GUIDE 57W

Work out what the machine would have made if it had run at its rating every hour of the year, then compare what it actually made with that. The ratio is the capacity factor, and it is the only number that plans anything. It already contains the hours the wind was too light, too strong or gone.

ASK 7W

Estimate the turbine's capacity factor on Vellan.

BEHIND THE BUTTON 159W

Why a rating is not a forecast. The rating is the output at the wind speed the machine is designed for, which on Vellan happens perhaps a fifth of the time. The rest of the year it produces less, and above a cut-out speed it produces nothing at all and turns itself out of the wind.

What a good capacity factor looks like. Onshore wind sits between about 25 and 35 per cent, offshore rather higher, and rooftop solar in this latitude around 10. A figure near 28 per cent on an exposed island is a genuinely good machine, and it is still nothing like the rating.

Why the brochure figure is not dishonest and is still wrong. It is a modelled output at a reference wind distribution, which Vellan does not have. The island is windier on average, and loses more hours to cut-out in winter gales. The only trustworthy figure is the one measured on this hill.

TAKES AS READ

the last full year of generation is representative of the wind here

VERDICT 102W

Two hundred and fifty kilowatts for 8,760 hours would be 2.19 million kilowatt-hours, and the machine made 613,000: a capacity factor of 0.28. That is a genuinely good number for onshore wind, and it is nothing like the rating. The rating is the output at one wind speed. The factor already contains every calm hour, every gale above cut-out and every hour of maintenance. Comparing against the brochure's 900,000 gives 0.68, which is a comparison between two estimates rather than a measurement. The 613,000 is what this hill actually produced, and it is the only figure a nine-year decision should rest on.

TAKEAWAY 15W

Capacity factor is what the machine actually gives, and it already contains the calm days.

12.2 Forty-one thousand, and what it saves

HARBOUR OFFICE · A PERSON · BALLPARK

SCENE 33W

The gearbox is £41,000. At a capacity factor of 0.28 the turbine displaces about 613,000 kilowatt-hours of generation a year. Diesel electricity costs the island about 7.4 pence a kilowatt-hour in fuel alone.

GUIDE 52W

Work out what a year of displaced diesel is worth, then divide the capital cost by it. That is the simple payback. Then look at what the number does not contain. Two of the things it leaves out point in opposite directions, and the council will be told only one of them.

ASK 8W

Estimate the simple payback time on the gearbox.

BEHIND THE BUTTON 149W

What simple payback leaves out. Maintenance on the machine. The cost of borrowing the £41,000. And the fact that the turbine cannot displace diesel at every hour it generates. Some of its output arrives when the cable is carrying the island anyway. All three make the real payback longer.

What it also leaves out in the other direction. The fuel price is not fixed. The cable is failing, and will be replaced at somebody's cost. The emissions saved are worth something in any future the council has to plan for. All three make the real case stronger.

Why simple payback is still the right first number. It is transparent, it can be checked by anybody in the room in one line, and both sides can then argue about the adjustments in the open. A discounted calculation with eleven assumptions in it wins a meeting and does not inform one.

TAKES AS READ

the fuel-only cost of diesel electricity is about 7.4 pence a kilowatt-hour

VERDICT 120W

Six hundred and thirteen thousand kilowatt-hours at 7.4 pence is £45,400 of fuel a year, against a £41,000 gearbox: a payback under twelve months. The striking part is not the gearbox. It is the two years the machine has already stood still. That is roughly £90,000 of diesel burnt waiting for a part, more than twice what the part costs. Simple payback leaves out three things that lengthen it. Maintenance, borrowing, and the hours the turbine generates while the cable is carrying the island anyway. It leaves out three things that shorten it. Fuel price risk, the failing cable, and the emissions. It is still the right first number, because anybody in the room can check it in one line.

TAKEAWAY 14W

Simple payback is capital over annual saving, and what it omits belongs beside it.

12.3 The sailing's own standing order

WATERWORKS · A ROOM · BALANCE

SCENE 38W

The second sailing would take on fresh water at the berth: about 6 cubic metres a crossing, twice a week through the season. Sorley has the recharge at 294,000 cubic metres a year and the licence at 340,000.

GUIDE 54W

Work out what the sailings would draw across a season. Set it against the two numbers that matter: the recharge, and the gap between recharge and licence. Then ask when it would be drawn, because a volume spread over a year and the same volume drawn in July are different demands on this store.

ASK 17W

Read what you need and report what the island actually takes out of the store each year.

BEHIND THE BUTTON 153W

Why the annual comparison understates it. Six cubic metres twice a week for a twenty-week season is 240 cubic metres, which is less than a thousandth of the recharge and looks like nothing. Every cubic metre of it is drawn in the twenty weeks when the store is lowest and the chloride is rising.

Why it is the marginal draw that counts. The island is already licensed to take more than the recharge, so any new demand is not being met out of spare capacity — there is none. It is met out of the store, which is the same as saying it is met out of next year.

What metering the ferry separately buys. It puts the sailing's own consumption on a line by itself, so a future council can see what the decision cost rather than inferring it from a total. That is why the condition matters more than the volume.

TAKES AS READ

the sailing takes on water at the berth rather than at the mainland end · carrying capacity, and what happens on the way past it — taken as read

VERDICT 137W

The term that does not announce itself is the leak. A fifth of everything the borehole lifts never reaches a tap. It is on no bill and no meter that anybody reads. It is the second largest withdrawal on the island. Counting only what is billed gives 280,000, and makes the recharge look nearly adequate. Counting the leak puts the total at 331,000 against a recharge of 294,000. That is the over-draw the July chloride has been reporting for five years. The ferry's own 240 cubic metres is the smallest number here. It lands entirely in the twenty weeks when the store is lowest. That is why it is metered separately rather than folded into a total. And the rain is the deposit rather than a withdrawal, which is the one line that must not be added.

TAKEAWAY 16W

A small volume drawn in the worst month is not a small demand on the store.

END OF THE DAY 14W

A generator is named for its best hour and paid for its average one.

What came in on the boat

BEFORE THE DAY — HUNT

Six standpipes, and one of them is the leak

The distribution main is losing about a fifth of what the borehole lifts, and Mairead Sorley, the water engineer, has narrowed it to the six standpipes along the island road. Find all six and read them. A fifth of the island's water is a bigger number than anything the ferry argument is about.

PLAN CARD 102W

Thursday. Hollis has found the same beetle in all four of his polytunnels. Nobody had seen it on Vellan before the spring the freight contract changed. Berhane makes the point Calloway cannot answer. Every extra sailing is an extra chance for something to arrive. The island has no inspection at the berth at all. The council pack also still needs one single emissions figure. Today you decide what makes this beetle a problem rather than a curiosity. You also put the island's two greenhouse sources on one line. An island is defended by its distance. A ferry is a bridge across it.

BRIEFING 14W

Something is eating the tunnel crops and it was not here three years ago.

WHAT YOU DECIDE 16W

Decide what an introduced species is doing and what the island's own emissions add up to.

13.1 Four conditions, and it has all of them

COMMON OFFICE · A ROOM · CHOICE

SCENE 39W

The beetle is in all four tunnels. It breeds three times a season and eats eleven of the crops Hollis grows. Nothing on the island has been seen eating it. It arrived within a year of the freight change.

GUIDE 61W

Being new is not the same as being invasive. Go through what the species does here rather than where it came from. How fast it reproduces. How broad its diet is. Whether anything controls it. And whether it can spread beyond where it landed. A species with one of those is a curiosity; one with all four is a management problem.

ASK 13W

Which condition makes this beetle a management problem rather than a new record?

BEHIND THE BUTTON 156W

Why "no natural predator" is the load-bearing condition. In its home range something eats it, and the population sits at a level that reflects that. Removed from its predators it does not sit anywhere. It grows until food or space stops it. That is why the same beetle is unremarkable at home and a problem here.

Why a broad diet matters so much on a small island. A specialist can only occupy what its host occupies. A generalist eating eleven crops can move between them as each is harvested, so there is no season in which its population has to crash.

Why the arrival route is the only part that can be closed. The beetle cannot be removed from four tunnels once it is breeding. What can change is the next arrival: inspection at the berth, cleaned pallets, and knowing which sailings carry soil and plants. A second sailing doubles the opportunities without doubling the inspection.

TAKES AS READ

the beetle was not recorded on the island before the freight change · island biogeography: area, isolation and the species a place can keep — taken as read

VERDICT 114W

All four conditions matter and one of them is doing the work. Where it comes from, a predator holds this beetle at a level nobody notices. Here nothing eats it. The population is limited only by food and space. On an island with eleven suitable crops, that limit is a long way off. Arriving from the mainland is how it got here rather than why it is a problem — most introductions establish nothing. Breeding three times a season is dangerous only because nothing is removing the offspring. And being found under cover is where it was noticed, not a constraint: the tunnels are warmer, and the crops outside are on the same list.

TAKEAWAY 15W

An invasive species is one that breeds fast, eats widely, is uncontrolled and can spread.

13.2 One figure, two gases

TIP AND SORTING YARD · A PERSON · BALLPARK

SCENE 35W

The pack needs one emissions figure. Seventy-eight thousand litres of diesel went into the sets this year and the cell vents about 14 tonnes of methane. The flare would burn the methane to carbon dioxide.

GUIDE 71W

Count the diesel from what went into the tank — litres times an emission factor — rather than from anything measured at the stack. Convert the methane to carbon dioxide equivalent, add the two and state the period the conversion belongs to. Then work out what the flare would leave. Burning methane does not remove the carbon. It changes which gas is carrying it, and that is most of the benefit.

ASK 10W

Estimate the island's total annual emissions in carbon dioxide equivalent.

BEHIND THE BUTTON 148W

What a flare actually does. It oxidises methane to carbon dioxide and water. The carbon still reaches the atmosphere, as a gas that traps a twenty-eighth as much heat per kilogram. So the climate effect falls by around ninety-five per cent, while the emitted mass barely changes.

Why diverting the compostables is better than flaring them. A flare deals with methane after it is made. Removing the organic matter means it is never made, and the same material becomes soil instead. The flare is what you fit for the material you cannot divert.

Why the period has to be on the line. At a hundred years methane is 28 times carbon dioxide, and the tip is about two thirds of the island's total. At twenty years it is 80 times, and the tip is five sixths of it. The same island, the same tonnages, two very different papers.

TAKES AS READ

the hundred-year potential is the basis the pack states

VERDICT 109W

Fourteen tonnes of methane at 28 is 392. Add the diesel's 205 and it gives about 600 tonnes of carbon dioxide equivalent. So two thirds of the island's climate footprint is a vent nobody meters. That is the sentence the pack has been missing. The flare is the striking part. Burning the methane leaves nearly the same mass of gas, and about 40 tonnes of equivalent. The carbon is now travelling as a gas that traps a twenty-eighth as much heat. Diverting the compostables is better still, because the methane is then never made. The period has to be stated: on the twenty-year basis the same tonnages give 1,325.

TAKEAWAY 15W

A flare changes which gas carries the carbon, and that is most of the benefit.

13.3 The distance, and the bridge across it

REEF STATION · A ROOM · CHOICE

SCENE 40W

Berhane's list of species recorded on Vellan for the first time runs at about one a year for thirty years. Four of them came in the three years since the freight contract changed. There is no inspection at the berth.

GUIDE 58W

An island's species list is held in place by two things: how big the island is and how hard it is to reach. The first cannot change. Ask what the second is worth in practice, and what a second sailing does to it. Then decide what the pack should ask for. The beetle already here cannot be removed.

ASK 12W

What should the pack ask for, given the beetle cannot be removed?

BEHIND THE BUTTON 150W

Why isolation matters as much as area. Arrivals are rare, so a species that goes locally extinct is not replaced, and a species that arrives faces no competitors that arrived recently either. Reduce the isolation and both halves change: more arrivals, and a community that has never had to hold its ground against newcomers.

What the numbers suggest and cannot prove. One new record a year for thirty years, and four in three, is a rate change worth acting on. It coincides with a freight contract rather than being explained by it. Berhane will say exactly that much and no more.

Why inspection is the cheap intervention. Nothing removes an established beetle from four polytunnels. What a berth inspection changes is the next arrival, at a cost of an hour a sailing. That is the whole argument for spending effort at the point of entry rather than on eradication afterwards.

TAKES AS READ

the first-record list has been kept by the same person throughout · species richness, and what an island holds that a mainland does not — taken as read

VERDICT 107W

Nothing removes a breeding beetle from four polytunnels, so effort spent on eradication buys very little. What can be changed is the next arrival. The rate change is what makes that urgent: one new record a year for thirty years, four in the last three. An hour's inspection a sailing is the cheapest environmental measure in the pack. A survey into how this beetle arrived is interesting and does not stop the next one; the route is already obvious enough to act on. And a ban on plant imports is stronger than the evidence needs, would stop Hollis restocking, and drives the same traffic into unrecorded channels.

TAKEAWAY 18W

An island is protected by its distance, and inspection is the only part of that anybody can restore.

END OF THE DAY 17W

Isolation is what an island's species are protected by, and a sailing is a hole in it.

DAY 14 OF 15

Nineteen on the register

PLAN CARD 127W

Friday. Liv Iversen, the schoolteacher, has brought the register to the office herself. The roll has fallen from 31 to 19 in twelve years. Two more leave for the mainland secondary school in August. Calloway's whole case rests on that number. Iversen will not let it be quoted as a slogan. It is an age structure, she says. An age structure is how many people there are at each age. It predicts the next decade whichever way the vote goes. Today you work out the island's own growth rate, and what that rate doubles or halves in. You also say what the age structure tells you the ferry cannot fix. A population is not a total. It is a shape, and the shape is what forecasts anything.

BRIEFING 12W

Iversen has the register and two of the nineteen leave in August.

WHAT YOU DECIDE 13W

Work out what the island's own population is doing, separately from the visitors.

14.1 Four flows, one stock

WATERWORKS · A ROOM · BALLPARK

SCENE 19W

Last year on Vellan: one birth, four people arrived, two died and seven left. The resident population is 91.

GUIDE 55W

A population changes by four flows and only four. Add what came in, take away what went out, divide by the population it happened to, and put it as a percentage. The sign is the answer to the day's question, and then the doubling-time arithmetic works the same way in reverse for a shrinking population.

ASK 8W

Estimate Vellan's population growth rate for the year.

BEHIND THE BUTTON 143W

Why all four flows have to be counted. Two of them are demographic and two are migration, and on a small island the migration terms are much the larger. A council discussing birth rates is discussing the smallest two numbers on the list.

What the rule of 70 does with a negative rate. It gives a halving time. At about minus four and a half per cent a year, ninety-one people halve in something like fifteen years. That is the number nobody in the room has said out loud. It is not affected by whether the school stays open next year.

Why the seven who left is the term to interrogate. It is the largest flow and the only one anybody can change. Why they left, and whether a second sailing changes those reasons, is the actual question hiding behind the ferry argument.

TAKES AS READ

the four flows are counted for the same year as the population figure · exponential growth, and reading a doubling time off a rate — taken as read

VERDICT 112W

Five in, nine out, over ninety-one, is minus 4.4 per cent a year. The rule of 70 works just as well downward. Seventy over four point four is about sixteen years to halve. Nobody in the ferry argument has said that out loud. The two migration terms dominate. One birth and two deaths, against four arrivals and seven departures. So a discussion about birth rates is a discussion about the smallest numbers on the sheet. The seven who left is the term worth interrogating, because it is the largest and the only one anybody can act on. The school roll is the consequence being argued about, not an input to the rate.

TAKEAWAY 16W

A population changes by four flows, and on a small island migration is the large pair.

14.2 A shape, not a total

HARBOUR OFFICE · A PERSON · CHOICE

SCENE 27W

Iversen's register: nineteen children, of whom two leave in August. The island's adults are heavily weighted above fifty-five, and four of the nine graziers are past seventy.

GUIDE 58W

Read the shape rather than the total. Ask how many people are of an age to have children here. Ask how many are of an age to stop working. Then ask what those two numbers do over ten years, without anybody moving. Then ask which of the ferry's promised effects would change the shape rather than the total.

ASK 12W

What would actually change the age structure rather than the summer total?

BEHIND THE BUTTON 159W

What an age structure predicts that a total cannot. The number of births in ten years is set by how many people of childbearing age are present now, and that is already determined. A total can be propped up by arrivals; the shape can only be changed by arrivals of a particular age.

Why an island's shape gets worse on its own. People leave in their late teens for education and work, and the ones who return, if any, return later. That removes the same cohort every year, so the structure hollows in the middle even when the total looks stable for a while.

What kind of arrival would actually change it. Families with young children, staying. Visitors change the summer total and nothing about the structure; seasonal workers change the summer total and the water demand. The ferry argument is about traffic, and the register is about residents, which is why the two keep talking past each other.

TAKES AS READ

the register and the age figures describe the same population · exponential growth, and reading a doubling time off a rate — taken as read

VERDICT 109W

The shape is what forecasts, and only one of these changes it. Families with children who stay add to both the young cohort and the working-age one, which is the two places the structure is hollow. Day visitors change nothing about residency at all. Seasonal workers change the summer total and the water demand, and leave in September. And a higher birth rate is not available as a lever. The number of people of childbearing age on Vellan is already fixed, and it is small. That is the thing Iversen keeps saying and the ferry argument keeps not hearing. One side is talking about traffic, the other about residents.

TAKEAWAY 18W

> A total can be propped up by arrivals; a shape changes only with arrivals of a particular age.

14.3 Ninety-one, or three hundred

TURBINE YARD · A ROOM · CHOICE

SCENE 35W

Ferris has to specify the replacement cable and the diesel sets for twenty-five years. The resident population is falling at four per cent a year; the July peak is set by visitors and is rising.

GUIDE 58W

Two trends run in opposite directions and the plant has to be specified against both. Work out which of them sets the plant and which sets the fuel, then say what the specification should follow. A falling total and a rising peak is not a contradiction; it is two different quantities, and only one of them buys equipment.

ASK 6W

What should the twenty-five-year specification follow?

BEHIND THE BUTTON 146W

Why the two diverge. Residents use energy all year and set the average. Visitors use it in twenty weeks and concentrate it in evenings. So they barely move the annual total. They land squarely on the peak half hour that sizes the plant.

What over-sizing costs. Diesel sets run inefficiently at part load, so a plant sized for a peak that rarely happens burns more fuel per unit all year. The cost of being wrong upward is continuous and invisible. The cost of being wrong downward is one dark evening, and very visible. That is why plants are usually over-sized.

Why the falling resident number does not save any equipment. Fewer residents lowers the average and the fuel bill. It does nothing to the July evening with the ferry alongside and the cable out, and that half hour is what the third diesel set exists for.

TAKES AS READ

the twenty-five-year specification cannot be revisited cheaply · energy against power: a kilowatt is not a kilowatt-hour — taken as read

VERDICT 116W

The two trends answer two different questions and the specification needs both. Plant is bought for the worst credible half hour. That is a July evening with the boat alongside and the cable out. It is rising, whatever the register does. Fuel, tariffs and anything a turbine displaces follow the annual total, which is falling with the resident population. Following the falling trend alone leaves the island dark on the one evening it has visitors. Following the peak for both over-sizes the sets. Diesels at part load burn more per unit all year, and that is a continuous invisible cost. The present average is stable because it averages the two trends into a number describing neither.

TAKEAWAY 13W

Two trends in opposite directions are two quantities, and each buys something different.

END OF THE DAY 13W

A population is an age structure, and its shape decides the next decade.

DAY 15 OF 15

What the island can carry

PLAN CARD 128W

Monday. The pack closes at four, and the vote is at seven in the chapel hall. Calloway has the landing fees and the school roll. She also has a ferry operator waiting on an answer. Berhane has 11 springs of transects and a rising July chloride reading. He has a beetle in four polytunnels as well. Sorley's recharge figure is the only number in the pack nobody disputes. Torr Vey is on the bench outside the hall and Marta is with him. Today you settle what the recommendation says. You settle what conditions it carries. You settle what the island is being asked to take on. A decision with conditions written into it is not the same thing as one without them. 91 people live on that difference.

BRIEFING 11W

The vote is at seven, and the pack closes at four.

WHAT YOU DECIDE 13W

Put one recommendation in front of the council, with the conditions it needs.

15.1 What it can carry, and for how long

HARBOUR OFFICE · A ROOM · CHOICE

SCENE 31W

The recommendation has to say whether the island can take a second sailing. The recharge is 294,000 cubic metres, the licence 340,000, and the store turns over in about nine years.

GUIDE 62W

Carrying capacity is not a number of people, it is a rate the place can sustain indefinitely. Ask what the binding constraint actually is here, then say what "for how long" means when the store buffers nine years of error. The answer is a condition rather than a yes or a no, and the condition has to name the quantity it caps.

ASK 6W

What should the recommendation actually say?

BEHIND THE BUTTON 140W

Why capacity is a rate and not a headcount. The same island supports more people using less water each, or fewer using more. What is fixed is the flow — 294,000 cubic metres of recharge a year — and everything else is a choice about how it is spent.

Why overshoot is possible at all. A stock lets a population live above the flow for as long as the stock lasts, which is what "eleven months of water in the ground" means. An overshoot is invisible while it is happening and expensive to reverse, on the same nine-year clock.

What a defensible recommendation looks like. It names the constraint, caps the rate at what the flow supports, states the period the cap is measured over, and says who measures it. Anything shorter than that is an opinion about a ferry.

TAKES AS READ

the recharge figure measured on day 2 stands as the pack's number · exponential growth, and reading a doubling time off a rate — taken as read

VERDICT 112W

The binding constraint is a rate: 294,000 cubic metres of recharge a year against a licence for 340,000. So the honest recommendation caps the rate. It does not answer yes or no about a boat. That form does the work both sides need: the sailing goes ahead, and it cannot be met out of the store, which is what "capped at recharge" means. Supporting it on the school roll ignores the constraint entirely. Opposing it treats a licence that can be rewritten as though it were the aquifer. Deferring for a survey spends eighteen months on a borehole that would draw from the same store, and the ferry contract will not wait.

TAKEAWAY 17W

Carrying capacity is a rate the place can sustain, and a stock lets you exceed it invisibly.

15.2 The sentence that has to come out

REEF STATION · A PERSON · PROTOCOL

SCENE 28W

Four sentences are in the draft pack. Dunmore has flagged one. Berhane has flagged a different one. The vote is in three hours and both flags are correct.

GUIDE 53W

Read each sentence against the evidence in the pack rather than against whether you agree with it. A sentence can be true, defensible and still wrong to include if the period is missing or the claim is stronger than the data. Find the two that fail and say which failure each one is.

ASK 14W

Match each draft sentence to what is wrong with it, or that nothing is.

BEHIND THE BUTTON 127W

The two ways a pack sentence fails. It claims more than the evidence supports, or it carries a figure with no period so it will be requoted as a permanent fact. Both are survivable in a conversation and neither survives being read out by somebody who disagrees.

Why the strongest sentence should stay in. Understating is a failure too: a paper that says less than it knows invites a decision the evidence would have prevented. The eleven-spring cover trend and the five-July chloride series both belong in at full strength.

Why this is the last stop of the fortnight. Everything measured over fourteen days arrives here as sentences somebody will read aloud at seven o'clock. The work is worth exactly as much as the sentences can bear.

TAKES AS READ

the pack's figures are the ones measured across the fortnight · concentration against load, and why a small pipe can matter — taken as read

VERDICT 113W

Two of these are the best evidence the island has, and belong in at full strength. A recharge figure measured this fortnight. And eleven springs of cover, read the same way each time. The chloride sentence is true and dangerous. The 191 is a July peak. Without its month it will be requoted as the island's water quality by the second meeting. The warming sentence is the one to cut. The correlation between the temperature series and the cover series can be stated. Causation needs control sites and bleaching tied to specific weeks, which nobody here has. Understating is also a failure, which is why the answer is not to soften all four.

TAKEAWAY 13W

A pack sentence fails by claiming too much or by dropping its period.

15.3 Rates, and the limits written down

COMMON OFFICE · A ROOM · SEQUENCE

SCENE 27W

Pike has the stocking limit, Okafor the volume plan, Ferris the gearbox and Sorley the cap. Four decisions, taken separately across a fortnight, going to one vote.

GUIDE 54W

Look for what the four have in common rather than what each says. Each one replaced something with something else of a different kind, and the same replacement happened four times. Name it, because that is what the council is being asked to adopt as a way of working rather than as four items.

ASK 13W

Put the fortnight's four replacements in the order the pack should present them.

BEHIND THE BUTTON 157W

What each of the four did. The stocking limit replaced a proportion of an uncounted total with a number of animals, inspected. The volume plan replaced tonnage with cubic metres per pound. The gearbox replaced a nameplate rating with a measured capacity factor. The cap replaced a licensed figure with a measured recharge.

Why that is one move and not four. In every case a figure was inherited from elsewhere: a template, a brochure, a proportion, an invoice. Each was replaced by a quantity measured on Vellan, with a period attached. The measurement was cheap in all four cases and nobody had taken it.

What it does not settle. The reef's decline is still not attributable. The beetle is still in the tunnels. The hole still has eleven years in it. The register is still falling at four per cent a year. A fortnight of measurement changes what the council can decide, not what the island is.

TAKES AS READ

the four decisions of the fortnight are the pack's substance

VERDICT 131W

The order is by how soon each constraint binds, which is the only ordering a council can act on in one evening. Water binds now: the licence exceeds the recharge, the store turns over in nine years and the ferry's own draw lands in the worst twenty weeks. The tip binds next, with eleven years of volume and a growth rate that shortens it. The turbine is a nine-month payback on a machine already bought, so it binds as soon as anybody signs. The common is the least urgent because the overstocking is recoverable within a season once it is counted. All four are the same move. An inherited number is replaced by a measured one, with a period attached. That is what the council is really being asked to adopt.

TAKEAWAY 16W

A measured local number with a period on it is worth more than an inherited figure.

END OF THE DAY 18W

A resource decision is a rate and a limit, and the conditions are where both are written down.

THE CLOSING CARD

The council carried the second sailing by five votes to four, with the abstraction licence capped at recharge — two hundred and ninety-four thousand cubic metres a year, metered at the berth and read monthly rather than annually. The west ground kept its quota at last season's landing rather than the tonnage the ferry paper had asked for. Nineteen children are still on the register, and the school has a second sailing's fees behind it for the first time in eleven years. One of the nineteen is Marta Vey, and the tap her class drinks from is read out at the council now rather than filed.

What it cost: the graziers accepted an inspected limit and two of them, Torr Vey among them, will not speak to the council office about it; the tip cell that should have been capped this summer waits for next year's money; and the diesel plant was sized on the July peak half hour, which means it runs at twenty-eight per cent of its plate for the other eleven months. What is unfinished: a fifth of everything the borehole lifts is still lost somewhere between it and the six standpipes, and nobody has dug up the island road to find out where. The tip is still up-catchment of the borehole, which is a decision made in 1974 and never revisited. The west ground is being worked at exactly the tonnage the model says it can replace, so one bad year is the whole margin. And eleven springs of one man counting the same reef is still the only baseline the island has.

The licence says what the rain puts back because you went and worked out what the rain puts back. You capped the abstraction at recharge instead of at what the borehole could lift, you priced the plant off the half hour in July that actually sizes it, and you found the fifth of the water that never arrives. Ninety-one people have an island to live on for another eleven years, and the number in the licence is yours.

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